

Mathematics Extension 1 Year 11

Further Transformations

Book 1

Version: 260928

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Learning Intention	Error! Bookmark not defined.

Syllabus Content

ME1-11-01 solves problems involving inequalities, functions and their inverses, graphical relationships between functions, and parametric equations

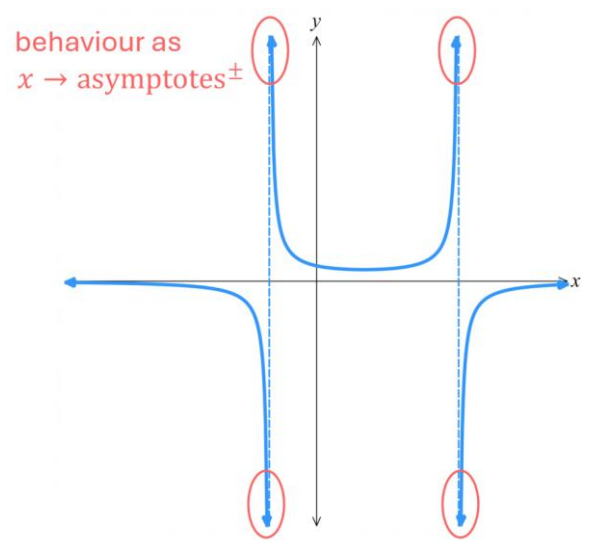
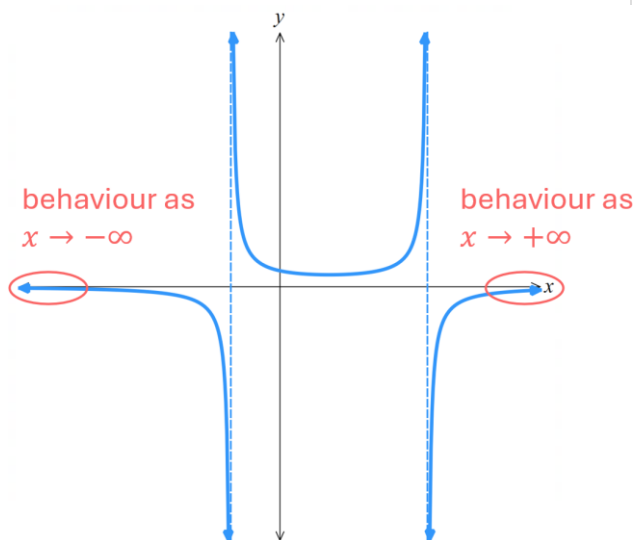
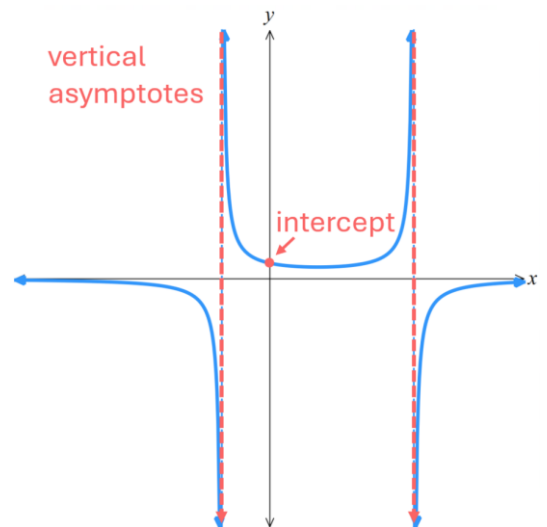
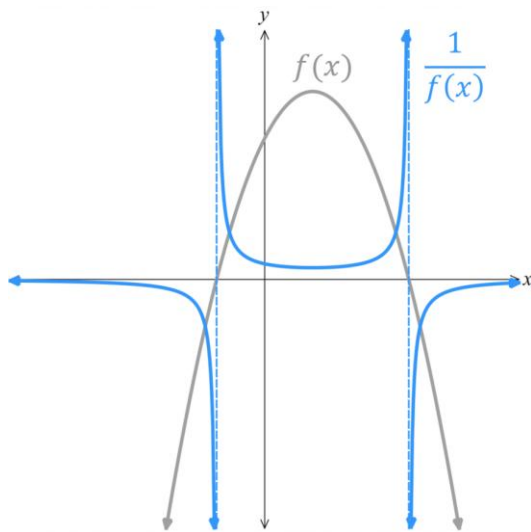
Graphical relationships

- Examine the relationship between the graph of $y = f(x)$ and the graph of $y = \frac{1}{f(x)}$ using graphing applications, and graph $y = \frac{1}{f(x)}$ given $y = f(x)$ in algebraic or graphical form, identifying any vertical and horizontal asymptotes of $y = f(x)$ and $y = \frac{1}{f(x)}$
- Graph $y = \sec x$, $y = \operatorname{cosec} x$ and $y = \cot x$ in both radians and degrees, identifying key properties including asymptotes, period, domain, range and symmetry, and compare each graph with the graph of its reciprocal
- Examine the relationship between the graph of $y = f(x)$ and the graphs of $y = |f(x)|$ and $y = f(|x|)$ using graphing applications, and graph $y = |f(x)|$ and $y = f(|x|)$ given $y = f(x)$ in algebraic or graphical form
- Examine the relationship between the graphs of $y = f(x)$ and $y = g(x)$ and the graphs of $y = f(x) + g(x)$ and $y = f(x) - g(x)$ using graphing applications, and graph $y = f(x) + g(x)$ and $y = f(x) - g(x)$ given $y = f(x)$ and $y = g(x)$ in algebraic or graphical form
- Determine the domains and ranges of the sum and difference of functions where possible, and, if appropriate, verify them using a graphing application
- Apply knowledge of graphical relationships to solve problems involving graphs of functions, justifying conclusions in the context of the problem where appropriate

Sketching The Reciprocal of a Function

① Reciprocal of a Function

- The reciprocal of a function $f(x)$ is $\frac{1}{f(x)}$, or $(f(x))^{-1}$.
- To sketch the graph, the key features to identify are **vertical asymptotes**, **intercepts**, **behaviour as $x \rightarrow \pm\infty$** and behaviour as $x \rightarrow \text{asymptotes}^\pm$.
- The function $y = \frac{1}{-(x-3)(x+1)}$ is shown below:



① Key Ideas for Sketching the Reciprocal of a Function

If $f(x) = 0$, then $\frac{1}{f(x)}$ is a vertical asymptote.

If $f(x) \rightarrow 0$, then $\frac{1}{f(x)} \rightarrow \infty$

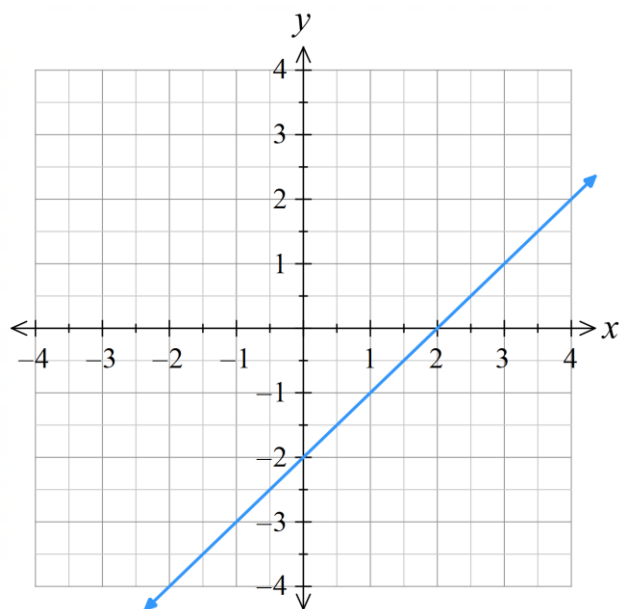
If $f(x) \rightarrow \infty$, then $\frac{1}{f(x)} \rightarrow 0$

If a point (a, b) is given, then the new point is $(a, \frac{1}{b})$

🔍 Sketching the Reciprocal of a Function

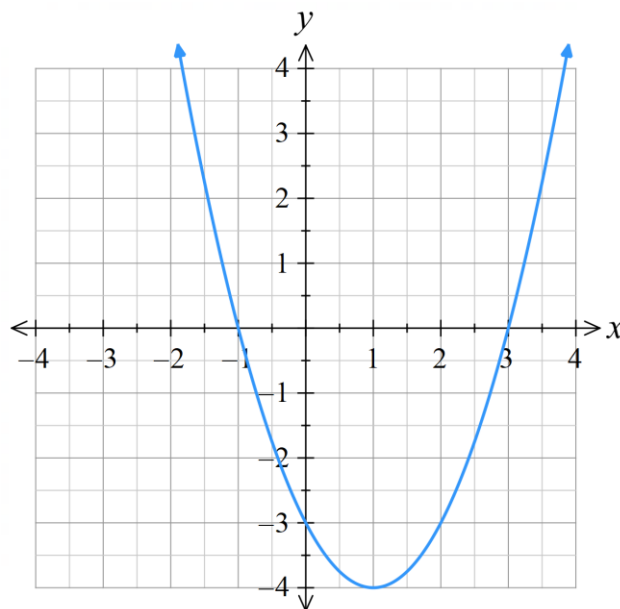
The graph of $y = x - 2$ is given.

Sketch $y = \frac{1}{x-2}$



The graph of $y = (x + 1)(x - 3)$ is given.

Sketch $y = \frac{1}{(x+1)(x-3)}$



① Rules for Sketching the Reciprocal of a Function

Use these rules to sketch $y = \frac{1}{f(x)}$ given only the graph of $y = f(x)$:

0. New coordinates and sign.

If the graph of $y = f(x)$ has a point (a, b) , where $b \neq 0$, then $y = \frac{1}{f(x)}$ has a point at $(a, \frac{1}{b})$.

If $f(x) = 1$, then $\frac{1}{f(x)} = 1$ as well.

1. Vertical asymptotes and behaviour.

If $f(x) = 0$, then $\frac{1}{f(x)}$ is a vertical asymptote.

If $f(x)$ is very small, then $\frac{1}{f(x)}$ is very large.

If $f(x) \rightarrow 0^-$, then $\frac{1}{f(x)} \rightarrow -\infty$

If $f(x) \rightarrow 0^+$, then $\frac{1}{f(x)} \rightarrow +\infty$

2. Horizontal asymptotes and behaviour.

If $f(x)$ is very large, then $\frac{1}{f(x)}$ is very small.

If $f(x) \rightarrow +\infty$, then $\frac{1}{f(x)} \rightarrow 0^+$ (horizontal asymptote)

If $f(x) \rightarrow -\infty$, then $\frac{1}{f(x)} \rightarrow 0^-$ (horizontal asymptote)

If $f(x)$ has an existing horizontal asymptote $f(x) \rightarrow c$, then $\frac{1}{f(x)} \rightarrow \frac{1}{c}$

3. Turning points:

If $f(x)$ has a turning point at (a, b) , then $\frac{1}{f(x)}$ has a turning point at $(a, \frac{1}{b})$.

4. Gradient:

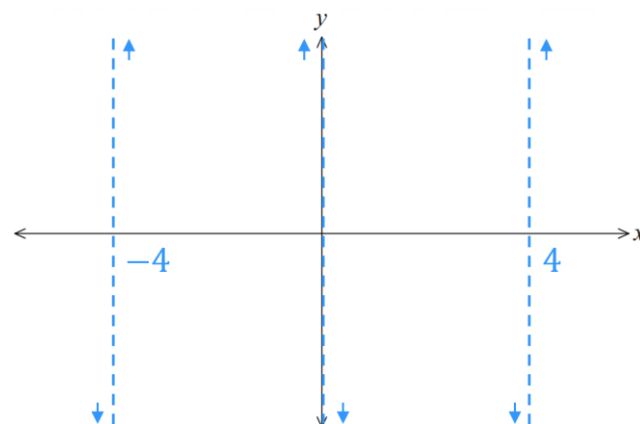
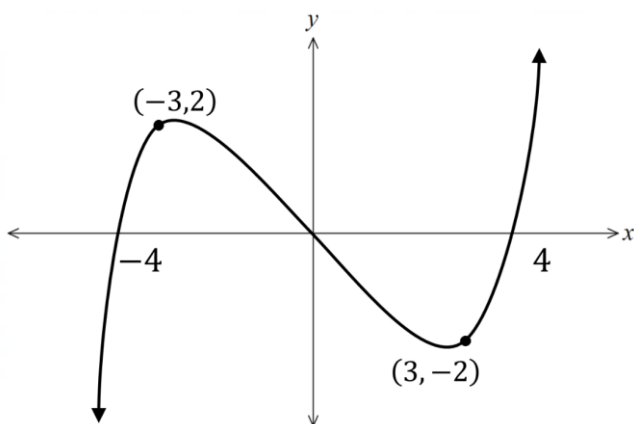
If $f(x)$ is increasing, then $\frac{1}{f(x)}$ is decreasing.

If $f(x)$ is decreasing, then $\frac{1}{f(x)}$ is increasing.

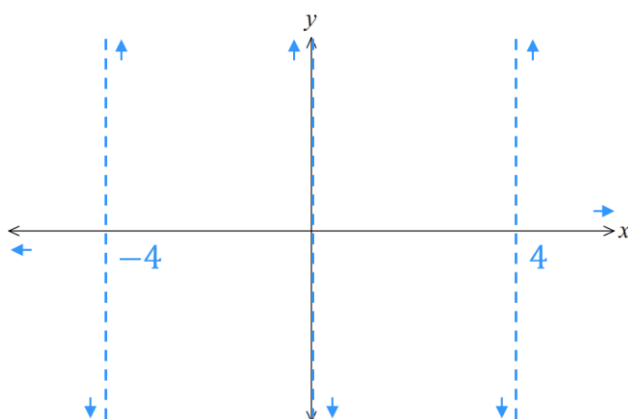
Example Sketch a function $\frac{1}{f(x)}$, given the graph of $f(x)$.

$y = f(x)$ is given. Sketch $y = \frac{1}{f(x)}$.

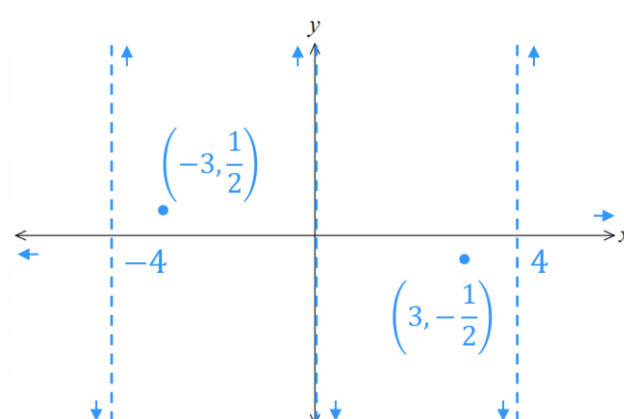
1. Vertical asymptotes & behaviour



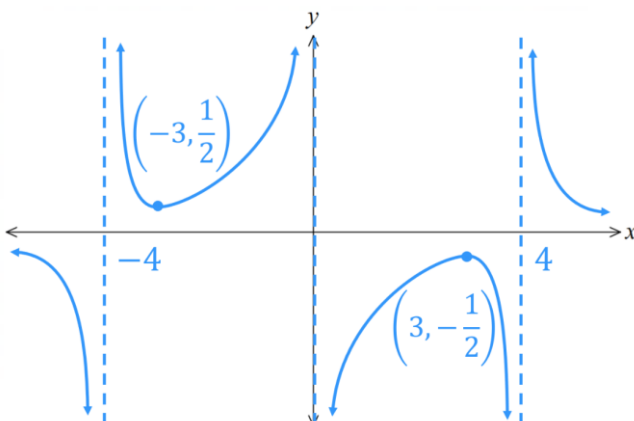
2. Horizontal asymptotes & behaviour



3. Turning points



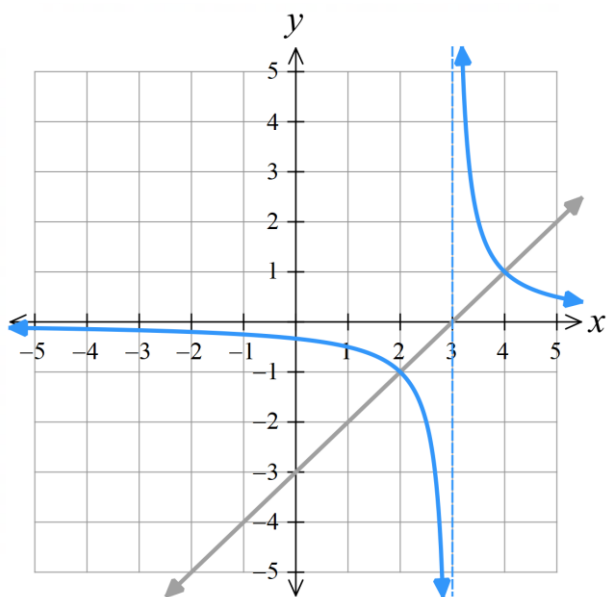
4. Gradient



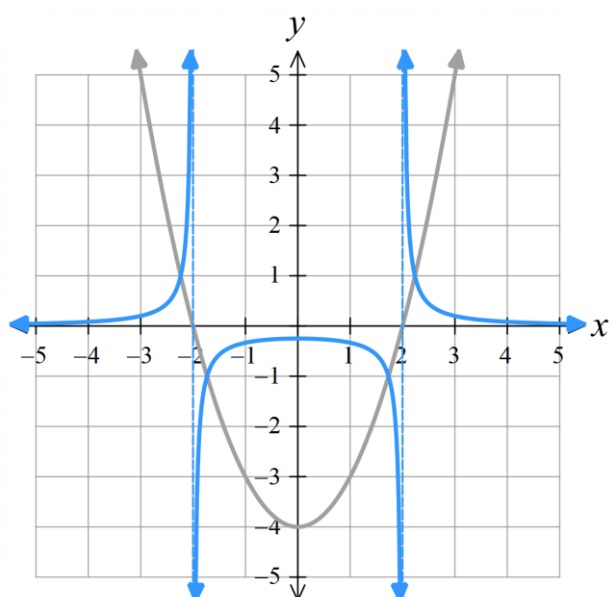
Guided Practice

$y = f(x)$ is given. Sketch $y = \frac{1}{f(x)}$ on the same axis.

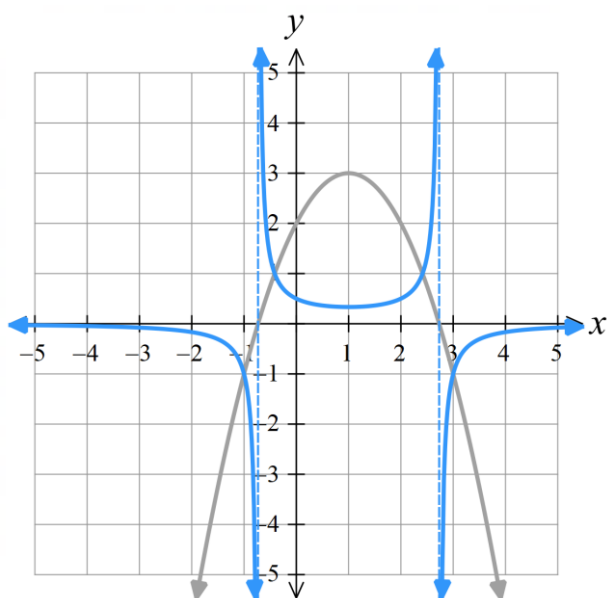
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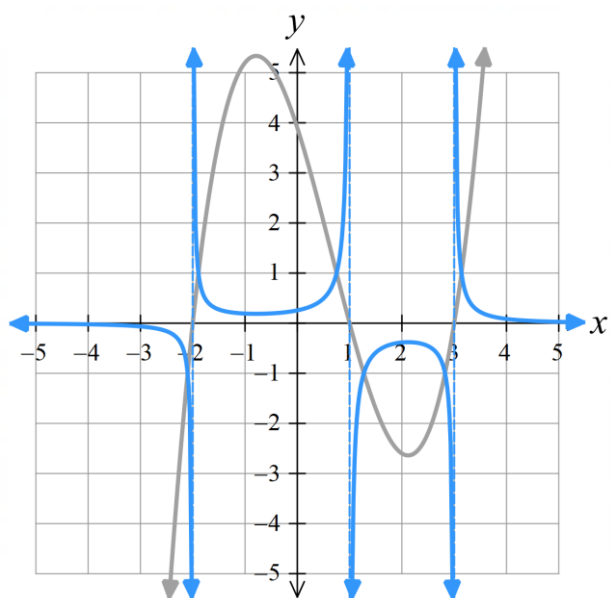
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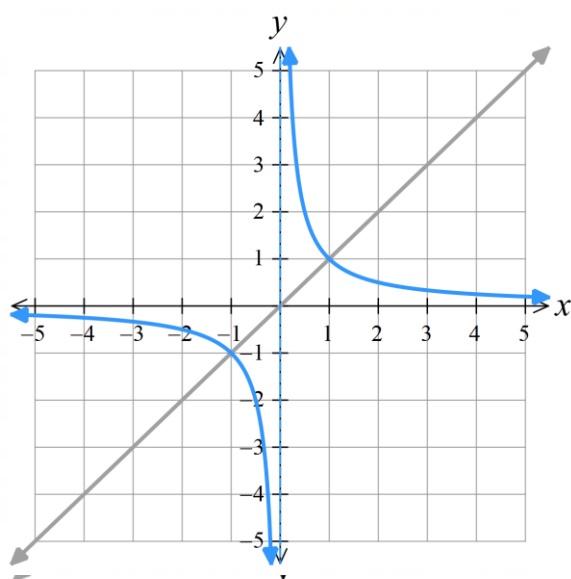


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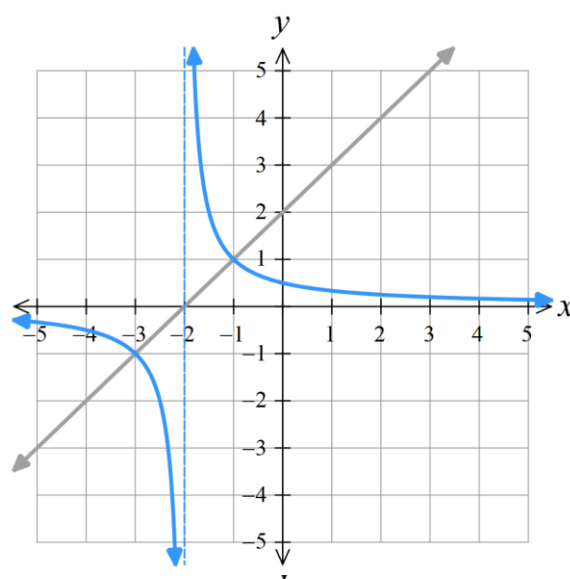


1 For each $f(x)$ given, sketch $\frac{1}{f(x)}$ on the same axis.

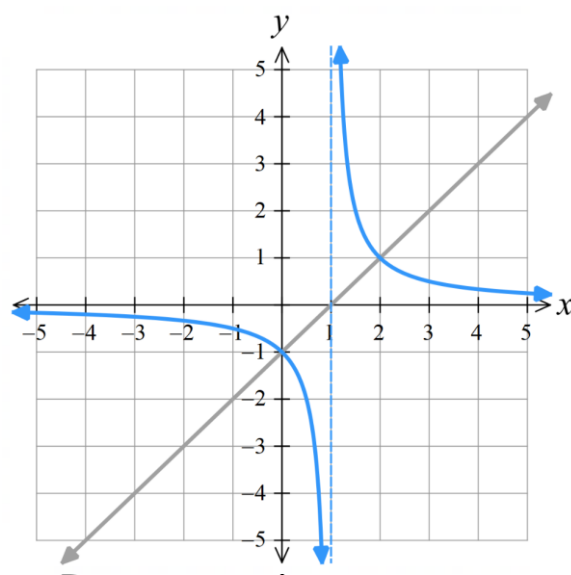
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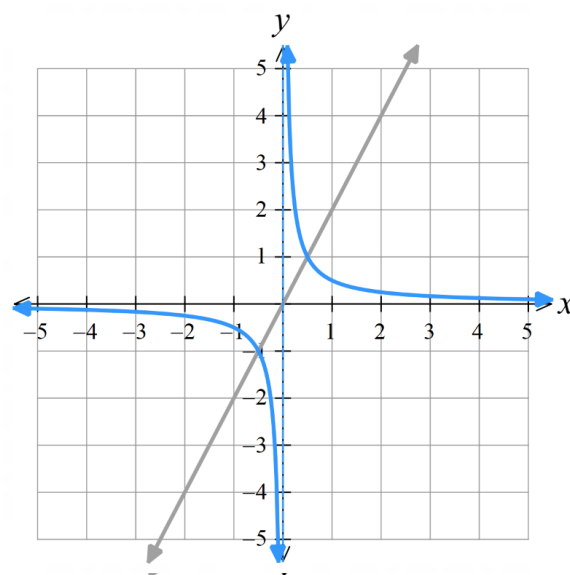
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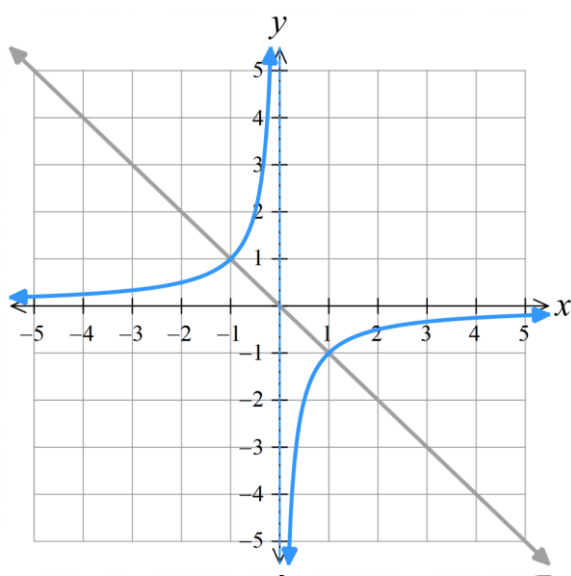
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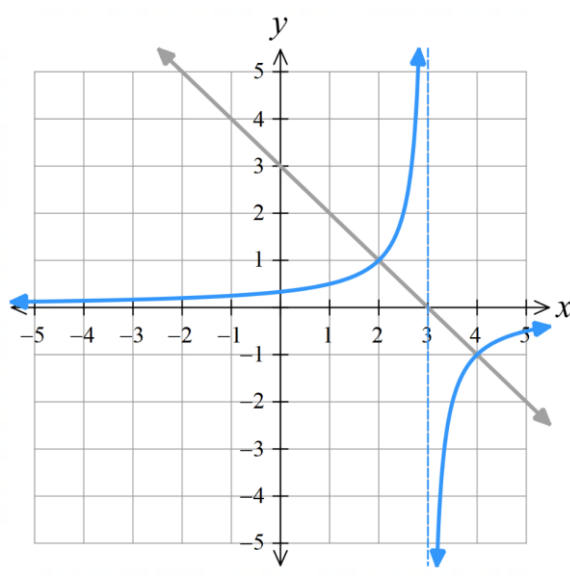
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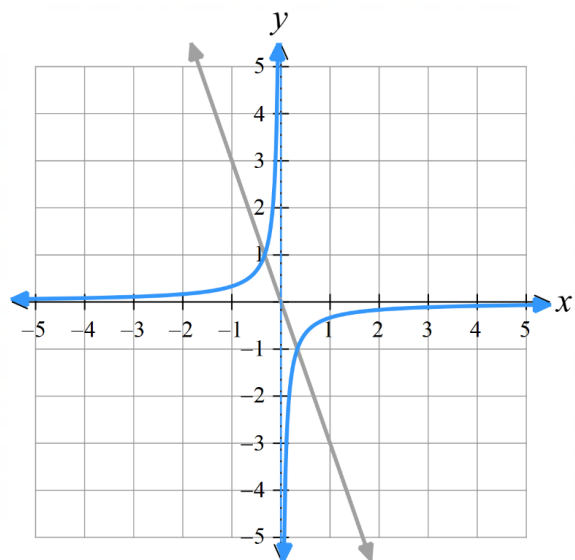
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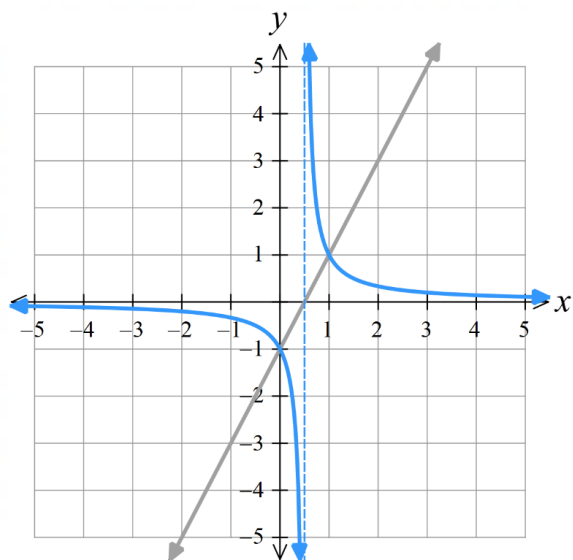
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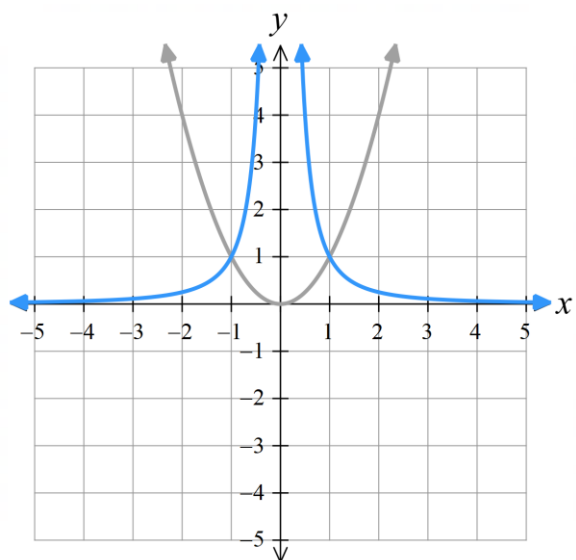
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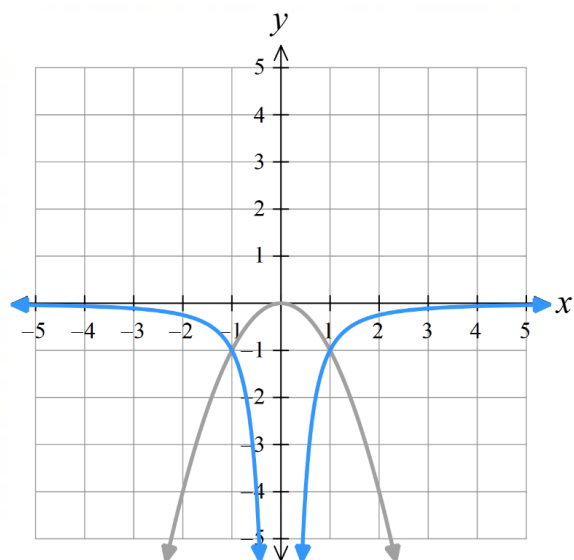
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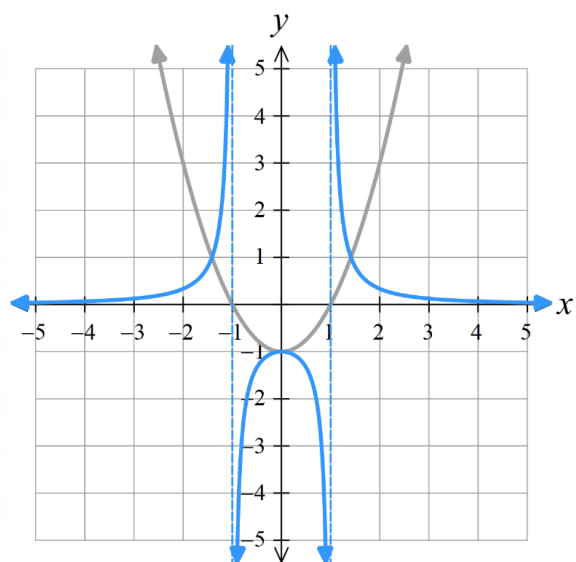
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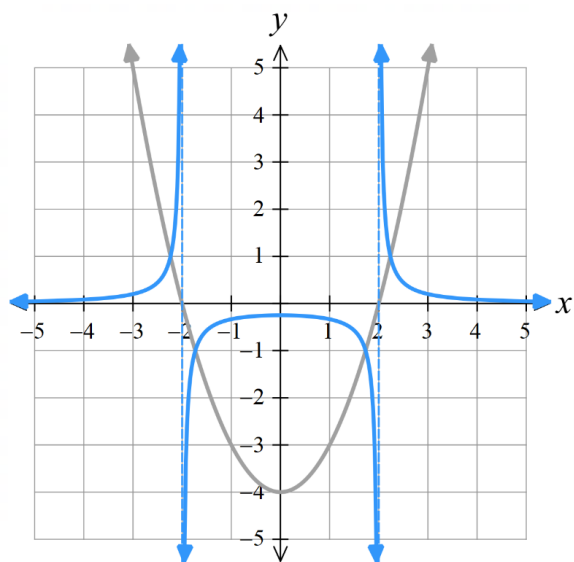
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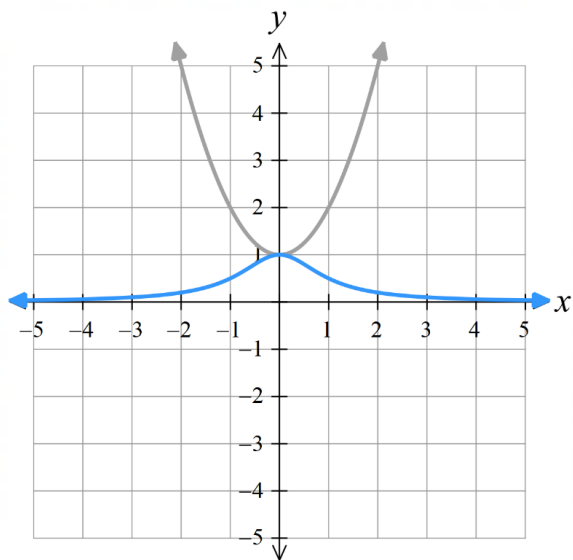
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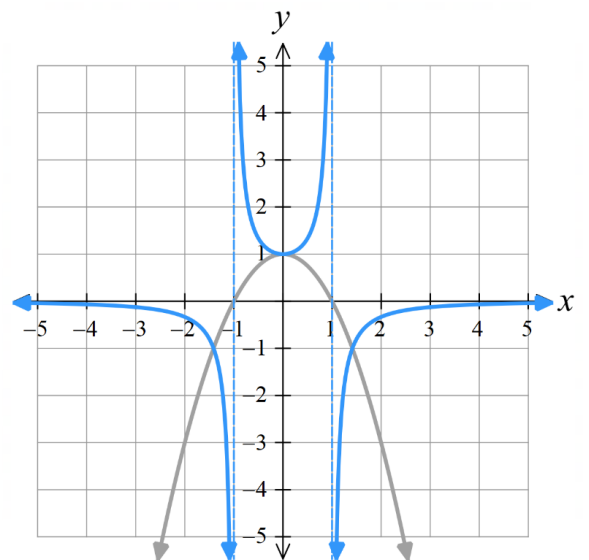
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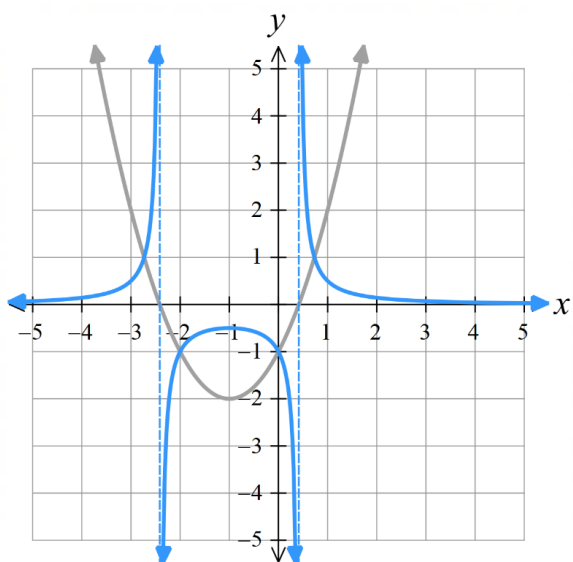
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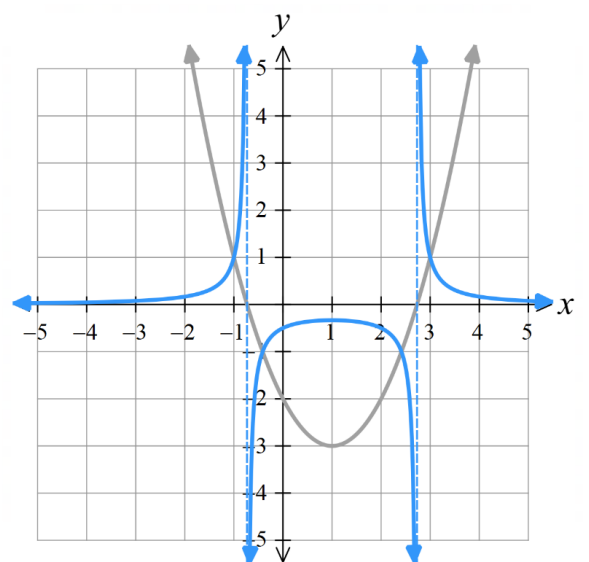
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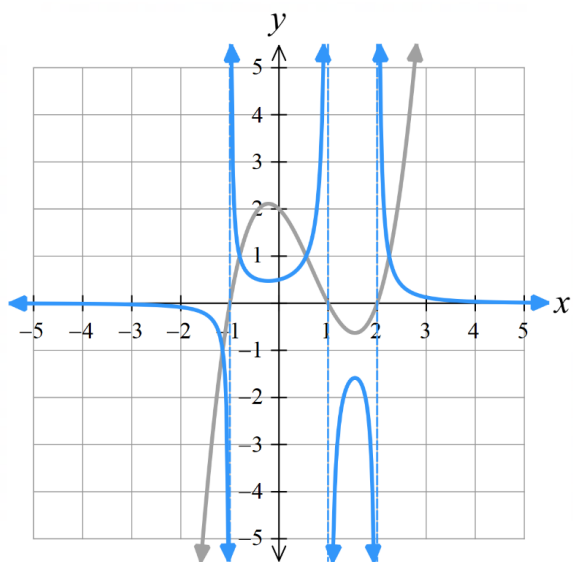
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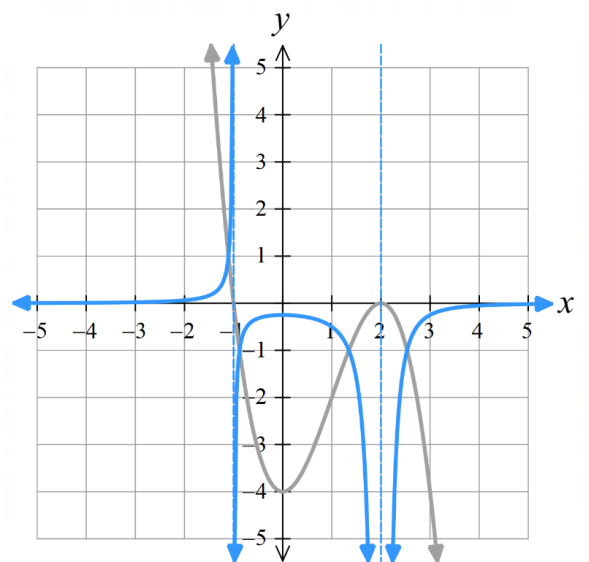
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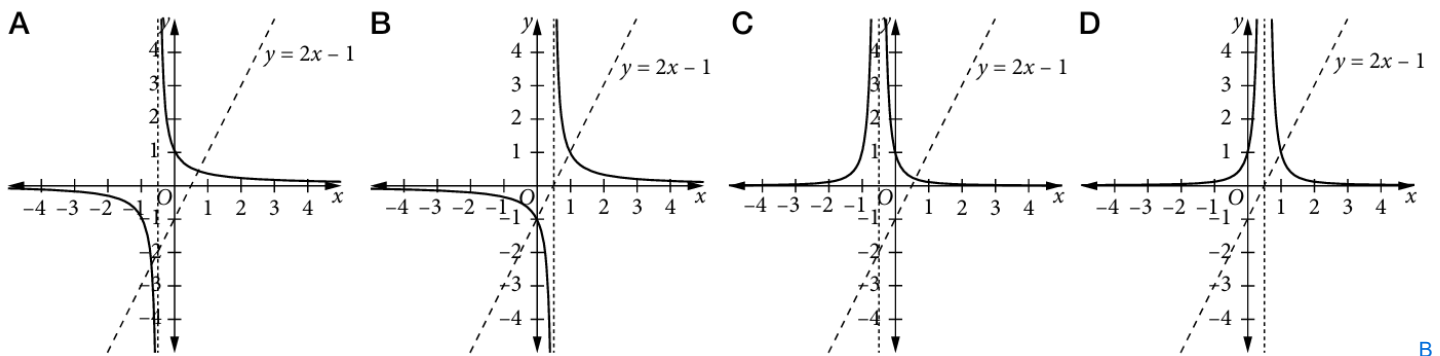
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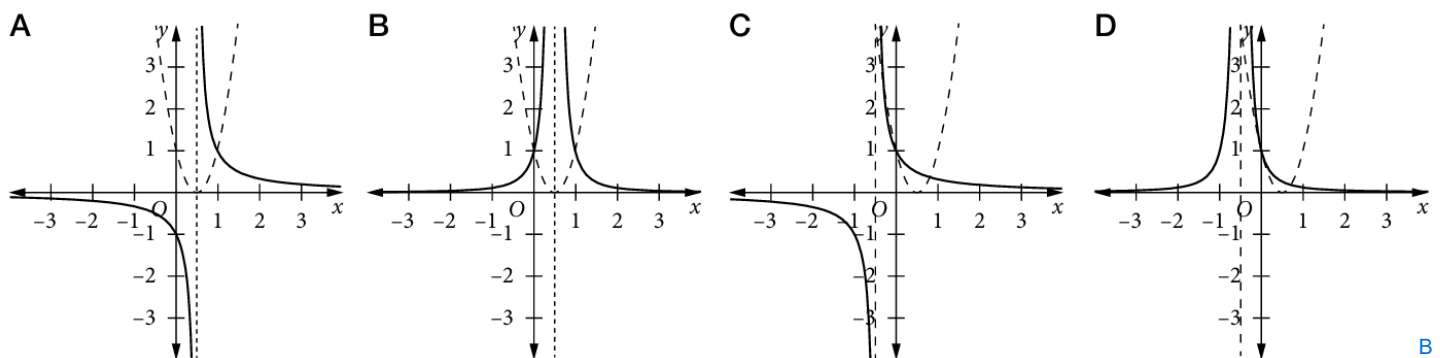


2 The graph of $y = 2x - 1$ is shown as a dotted line. Which graph represents $y = \frac{1}{2x-1}$?



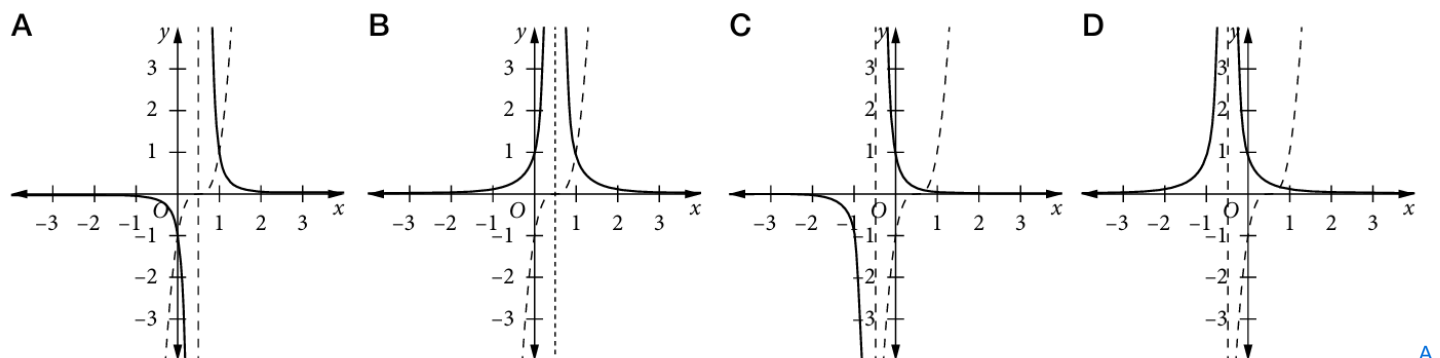
B

3 The graph of $y = (2x - 1)^2$ is shown as a dotted line. Which graph represents $y = \frac{1}{(2x-1)^2}$?



B

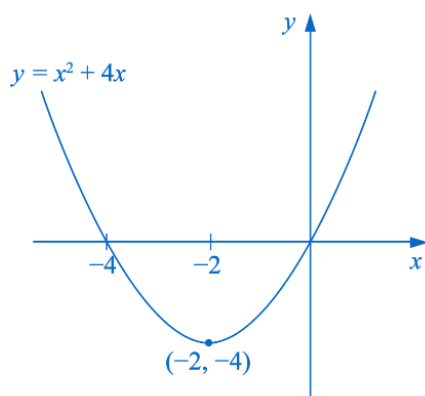
4 The graph of $y = (2x - 1)^3$ is shown as a dotted line. Which graph represents $y = \frac{1}{(2x-1)^3}$?



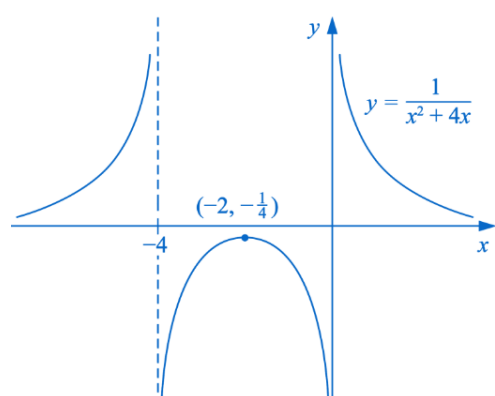
A

5 2010 HSC Extension 2 Band 2

a Sketch the graph of $y = x^2 + 4x$

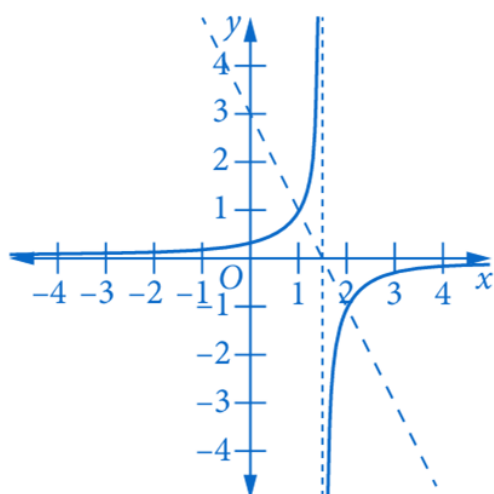


b Sketch the graph of $y = \frac{1}{x^2 + 4x}$

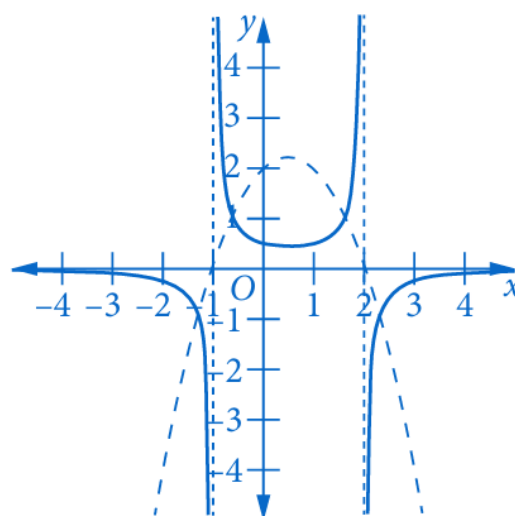


6 Sketch the reciprocals of these functions, showing all significant features.

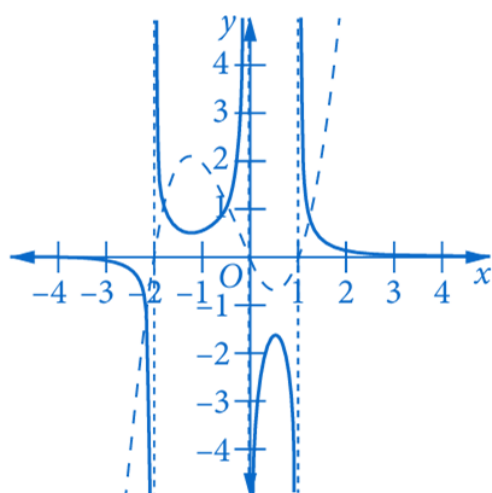
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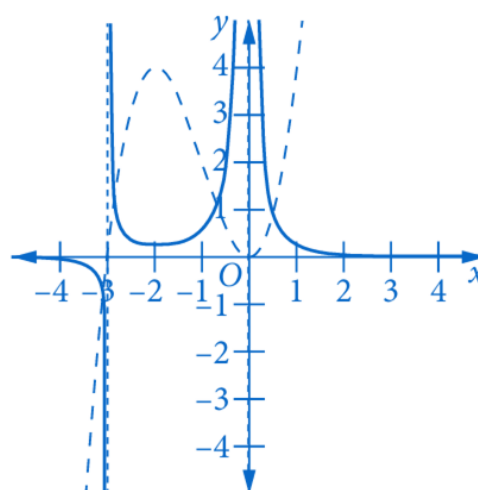
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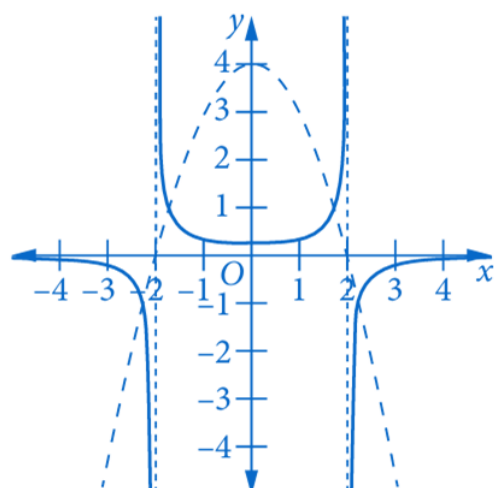
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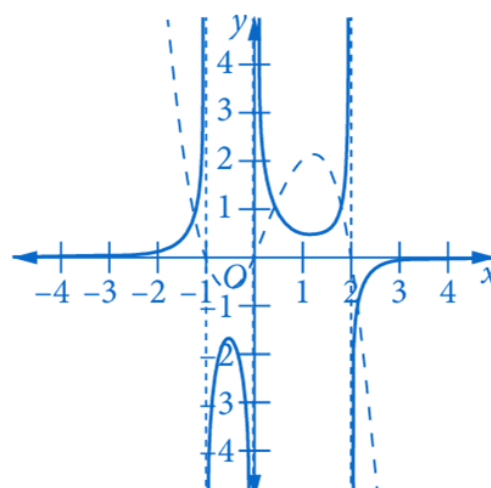
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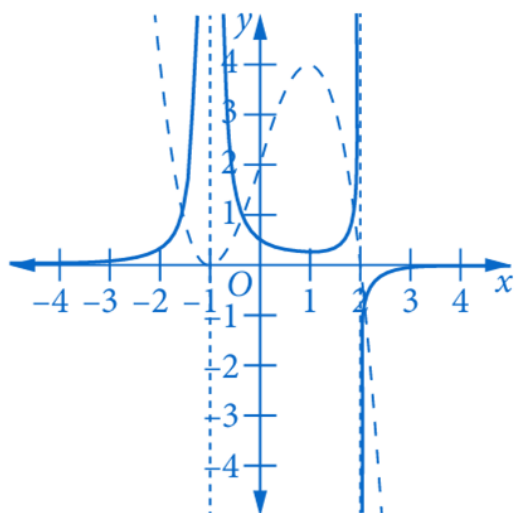
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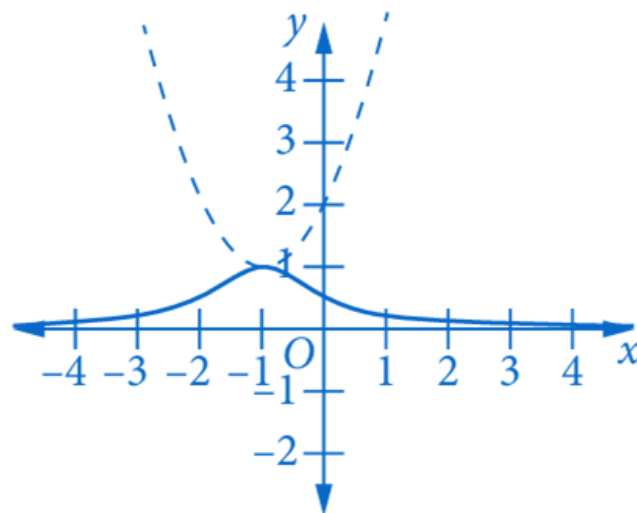
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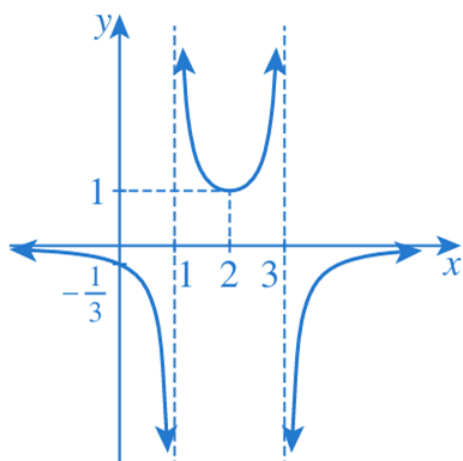


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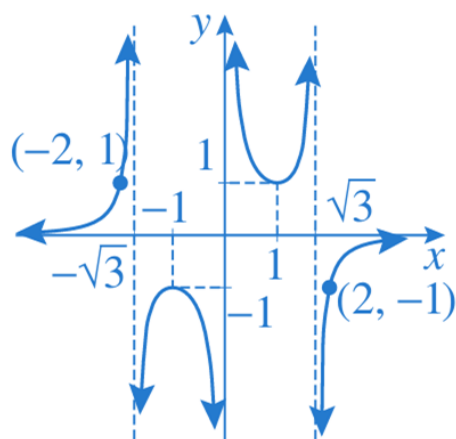


7 Sketch the reciprocal of these functions, showing all important features.

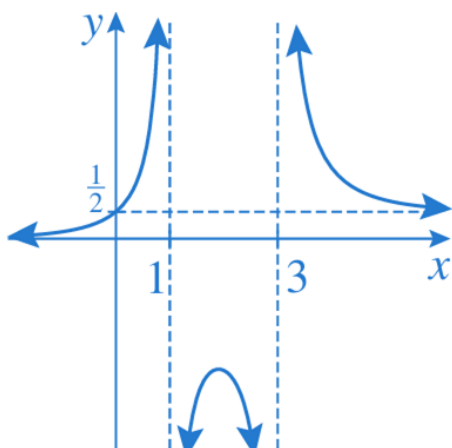
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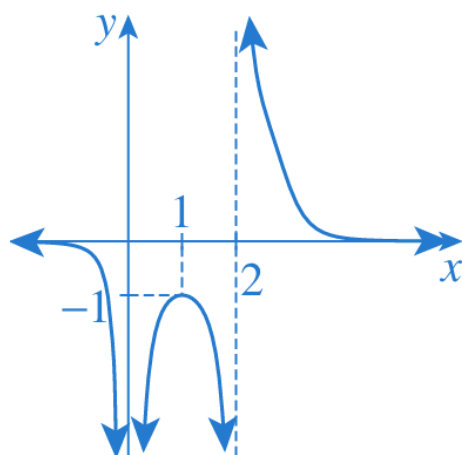
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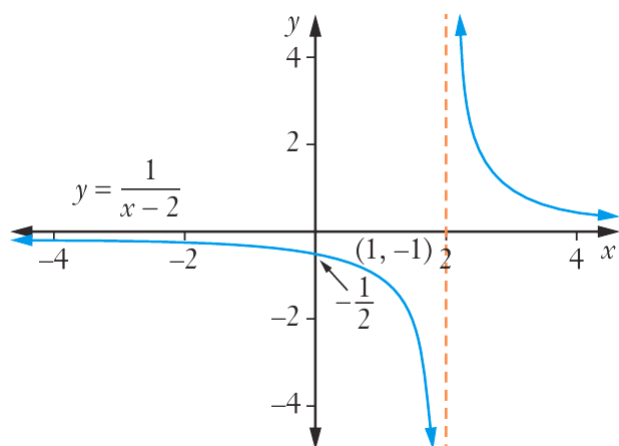
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8 By sketching an appropriate “base” function and reciprocating, sketch these equations.

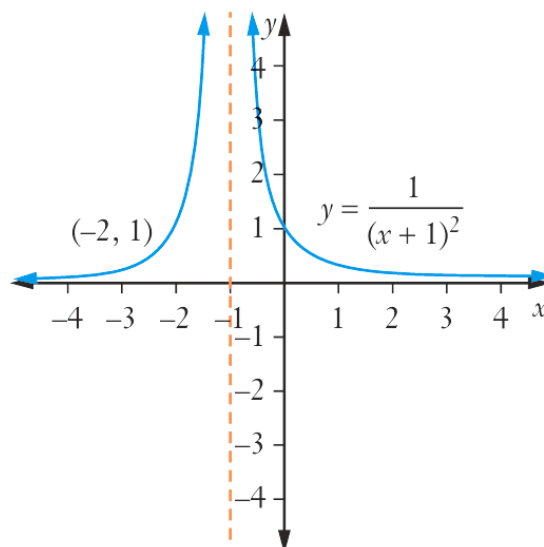
a

$$y = \frac{1}{x-2}$$



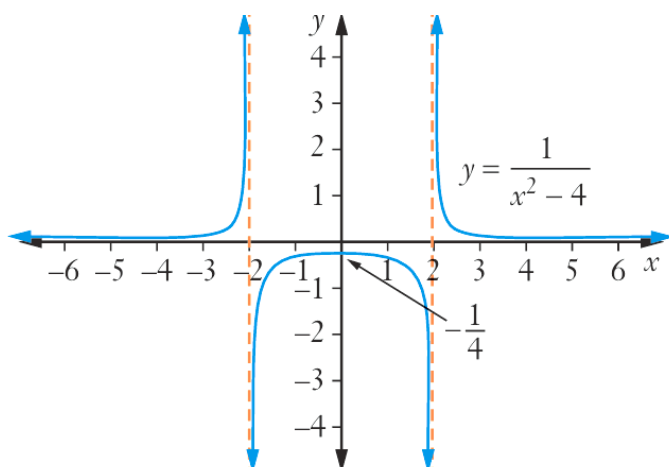
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$$y = \frac{1}{(x+1)^2}$$



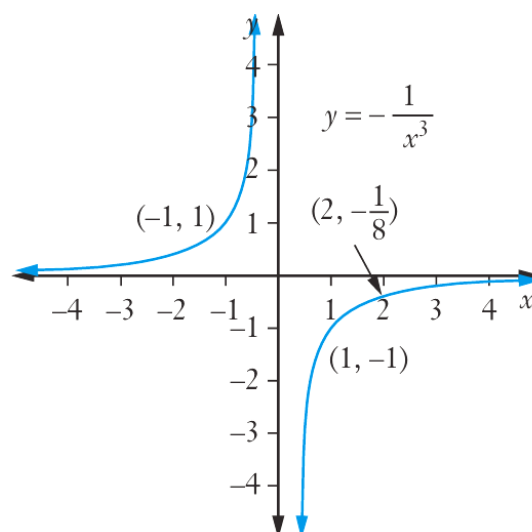
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$$y = \frac{2}{x^2 - 4}$$



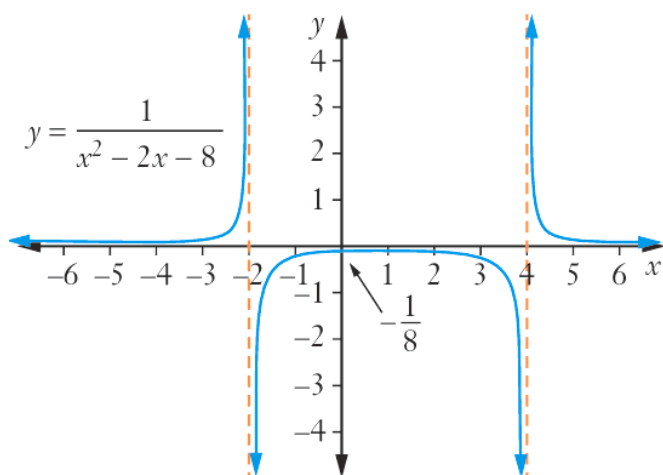
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$$y = -\frac{1}{x^3}$$

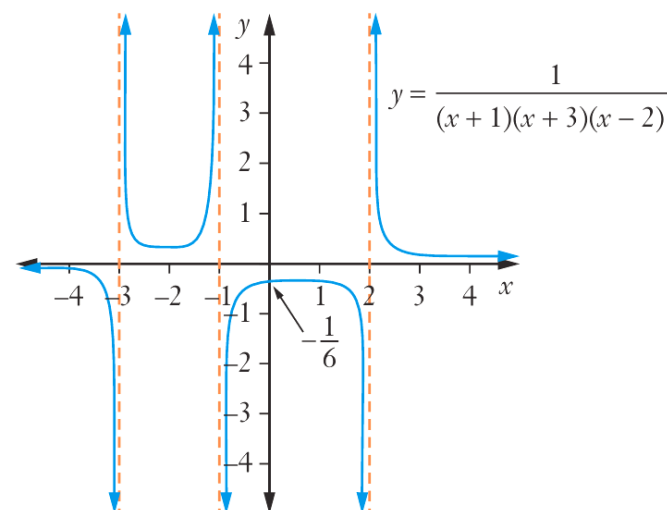


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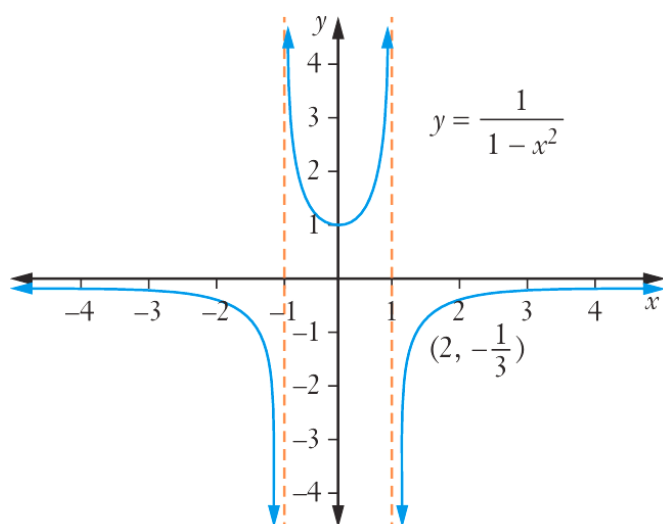
$$y = \frac{1}{x^2 - 2x - 8}$$

**f**

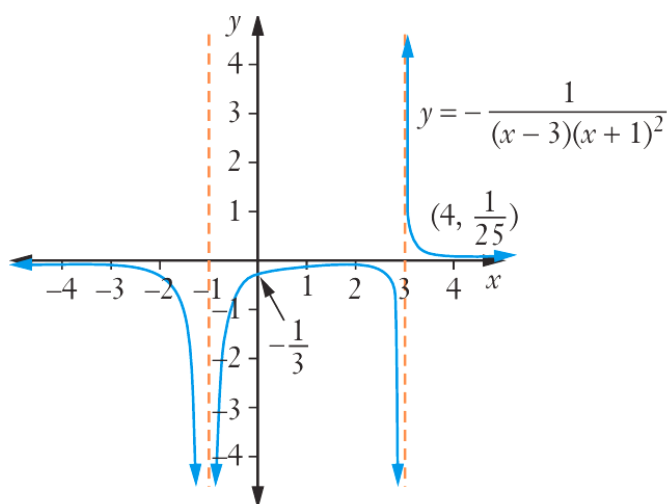
$$y = \frac{1}{(x+1)(x+3)(x-2)}$$

**g**

$$y = \frac{1}{1-x^2}$$

**h**

$$y = \frac{1}{(x-3)(x+1)^2}$$



- 9 $f(x) = \frac{1}{x+1}$, Grace was asked to find the equation and domain of $g(x) = \frac{1}{f(x)}$. She says:



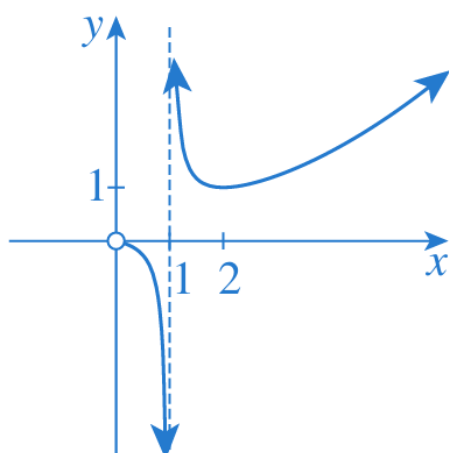
$\frac{1}{f(x)}$ would have an equation of $x + 1$, this has domain $x \in \mathbb{R}$

Explain why her equation is correct but domain is incorrect.

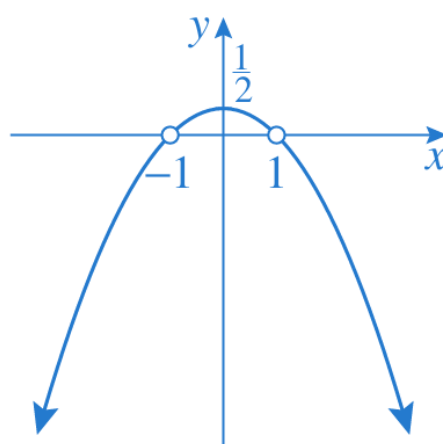
The original function is undefined for $x = -1$, so the new function is $g(x) = x + 1$, similarly undefined for $x \neq -1$.

- 10 The graph of $y = f(x)$ is given. Sketch $y = \frac{1}{f(x)}$

a

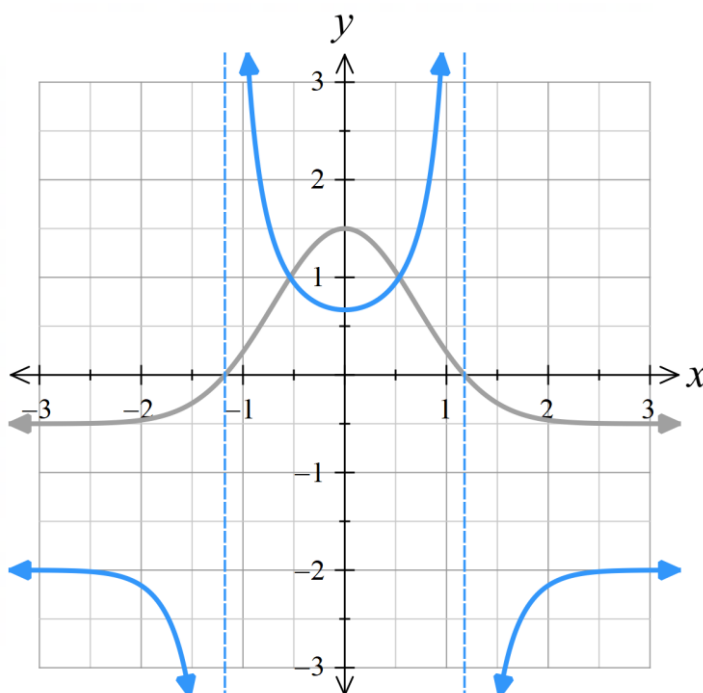


b



11 HSC Sample Question Band 4

The function $y = f(x)$ is given. Sketch $y = \frac{1}{f(x)}$ and state the range in interval notation.



$$R: (-\infty, -2) \cup \left[\frac{2}{3}, \infty\right)$$

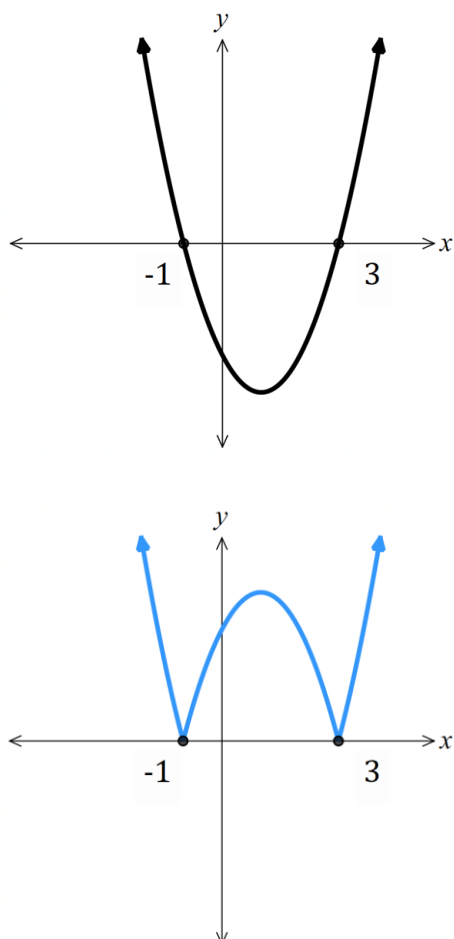
Graph of the Absolute Value of a Function

① Absolute Value Transformations

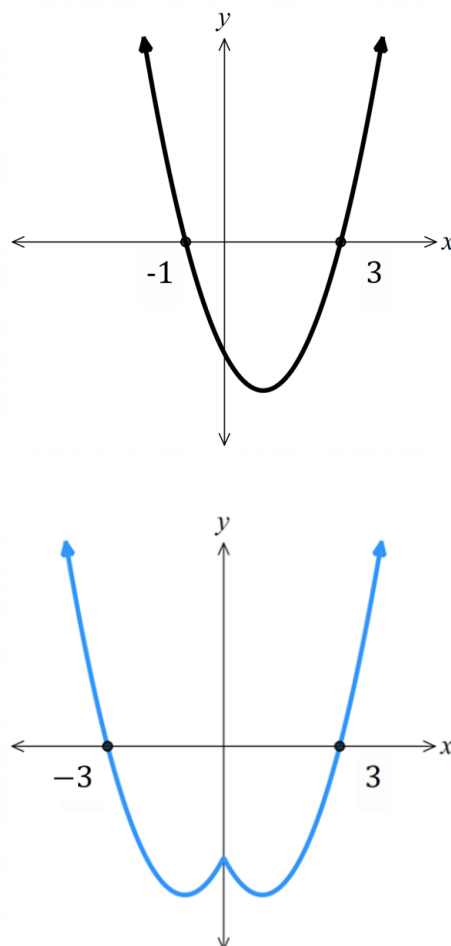
$y = f(x) $	$y = f(x)$
1. All points on and above the x-axis remain unchanged. 2. Reflect all points below the x-axis upward.	1. All points on and to the right of the y-axis remain unchanged. 2. Remove all points left of the y-axis and reflect all points right of the y-axis across.

Example Sketch a function $y = |f(x)|$, and $y = f(|x|)$ given the graph of $f(x)$.

$y = f(x)$ is given. Sketch $y = |f(x)|$.



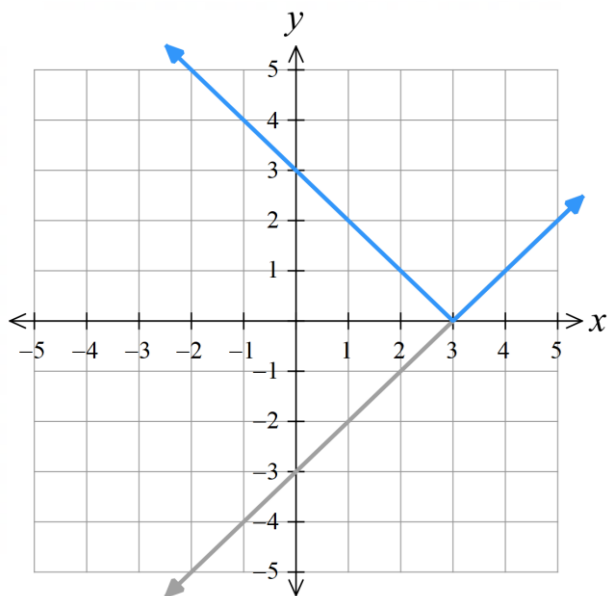
$y = f(x)$ is given. Sketch $y = f(|x|)$.



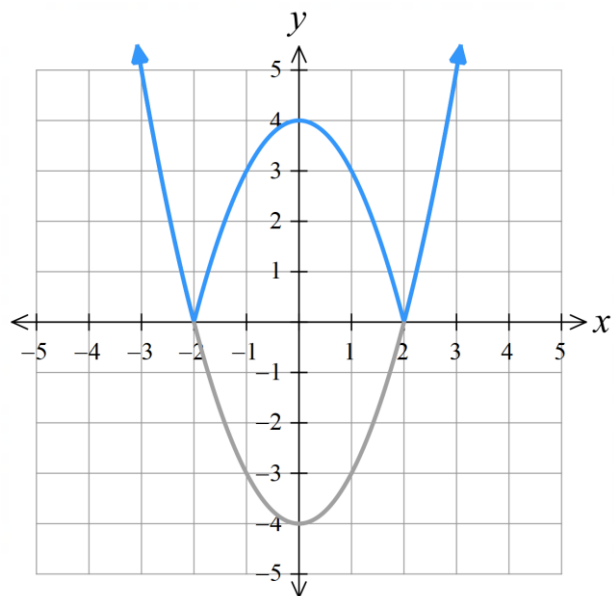
Guided Practice

$y = f(x)$ is given. Sketch $y = |f(x)|$ on the same axis.

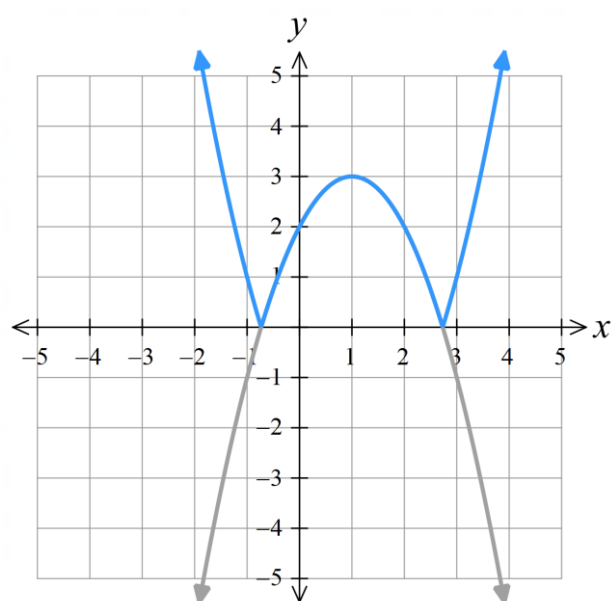
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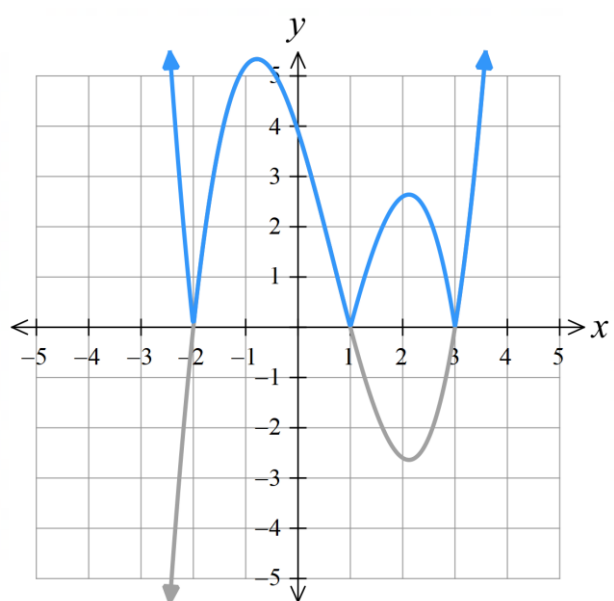
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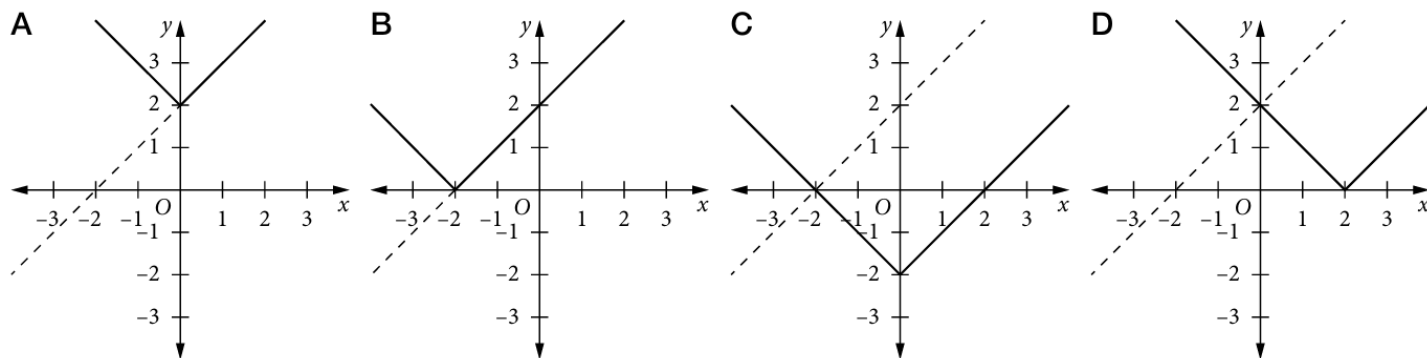
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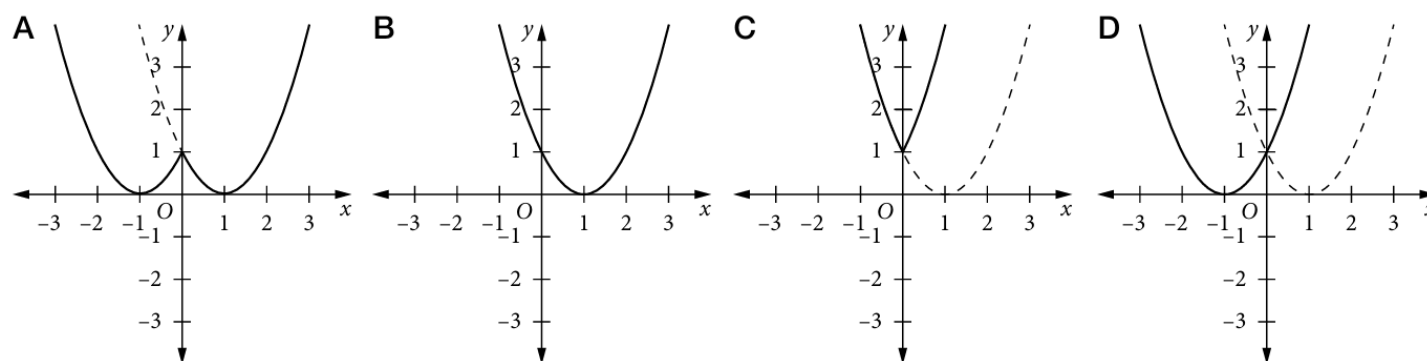


1 The graph of $f(x) = x - 2$ is shown as a dotted line. Which graph represents $y = |x - 2|$?



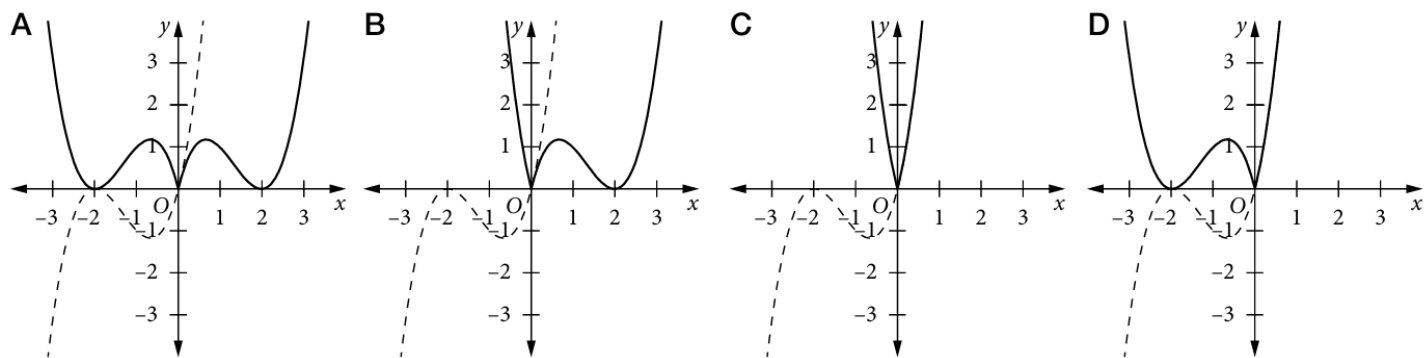
B

2 The graph of $f(x) = (x - 1)^2$ is shown as a dotted line. Which graph represents $y = (|x| - 1)^2$?



A

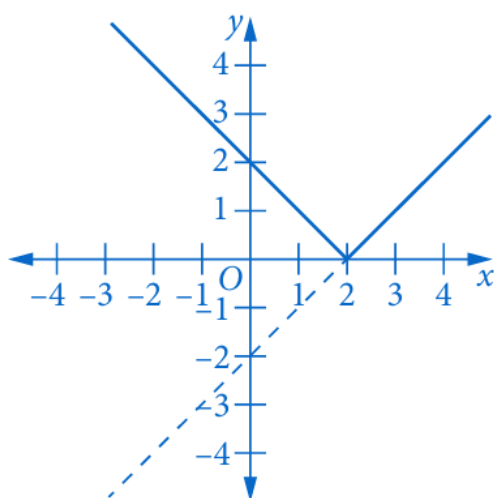
3 The graph of $f(x) = x(x + 2)^2$ is shown as a dotted line. Which represents $y = |x|(|x| + 2)^2$?



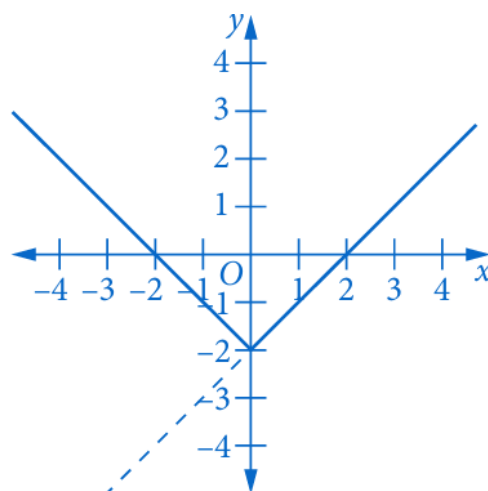
C

4 The graph of $f(x)$ is given,

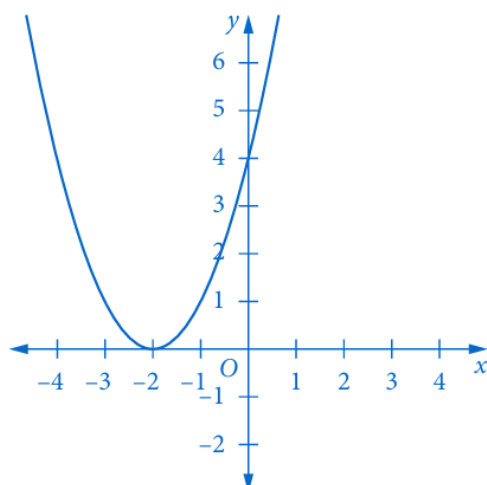
a Sketch $y = |f(x)|$



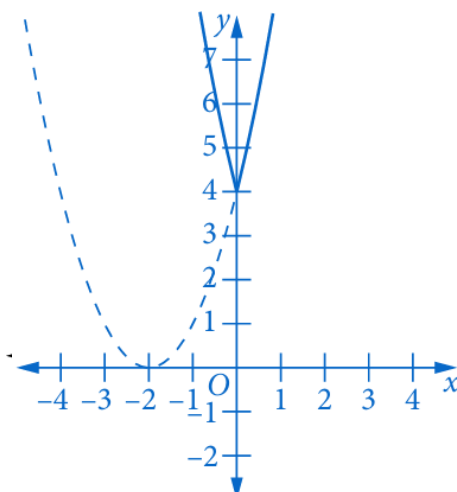
b Sketch $y = f(|x|)$



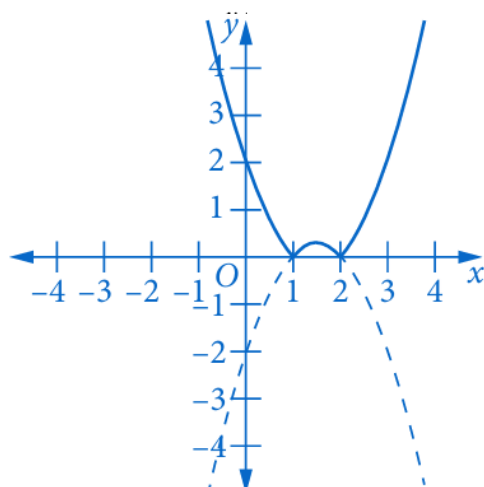
c Sketch $y = |f(x)|$



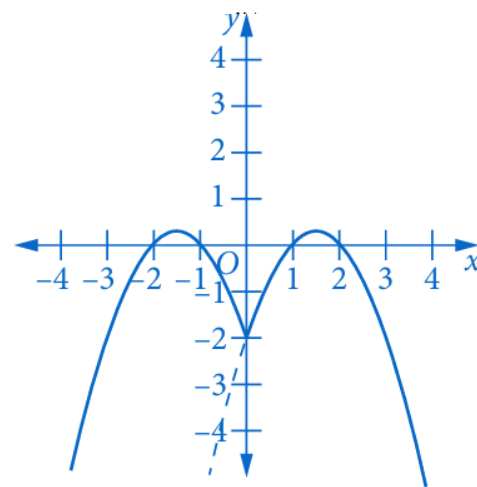
d Sketch $y = f(|x|)$



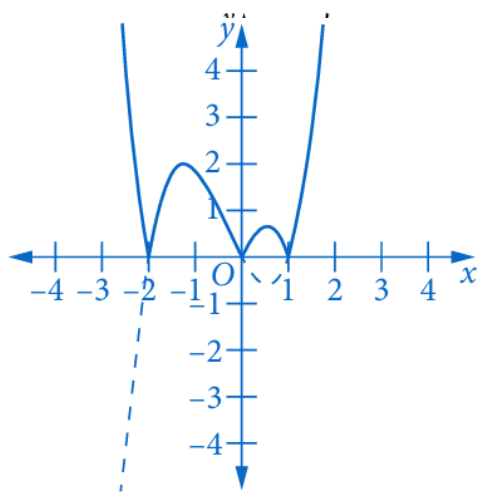
e Sketch $y = |f(x)|$



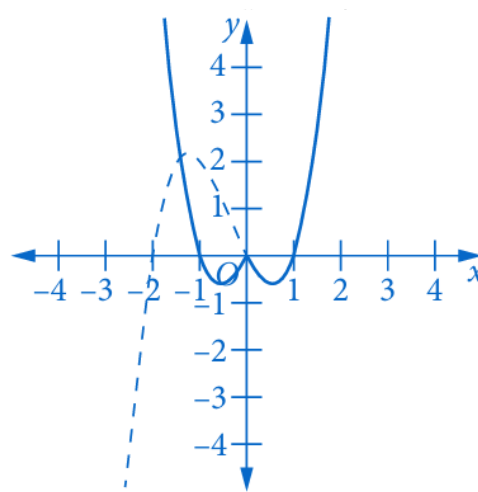
f Sketch $y = f(|x|)$



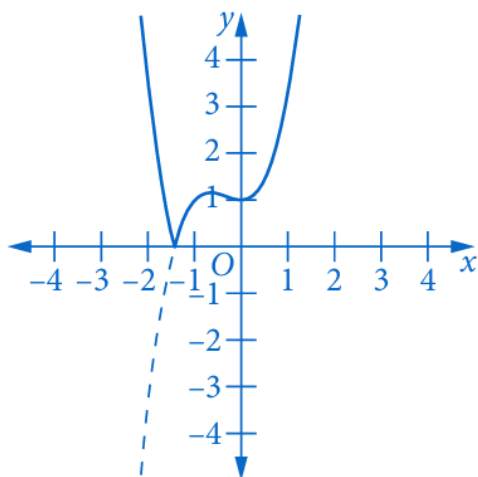
g Sketch $y = |f(x)|$



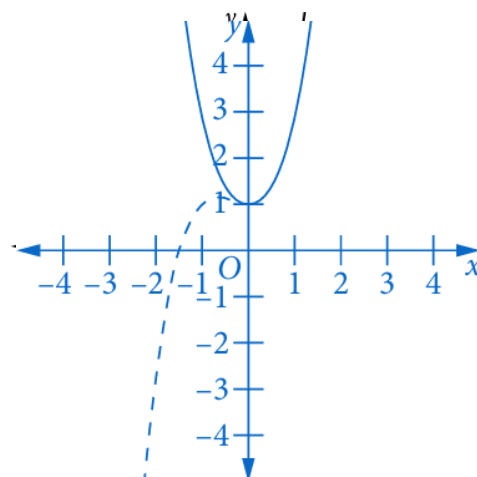
h Sketch $y = f(|x|)$



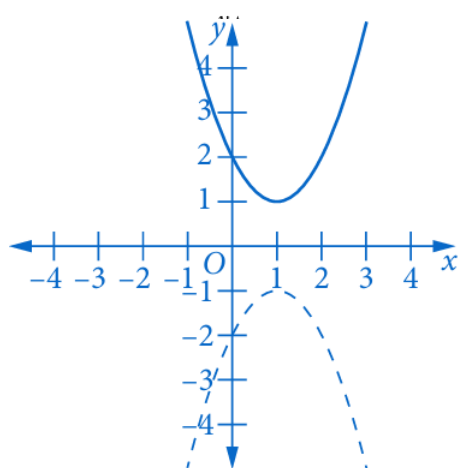
i Sketch $y = |f(x)|$



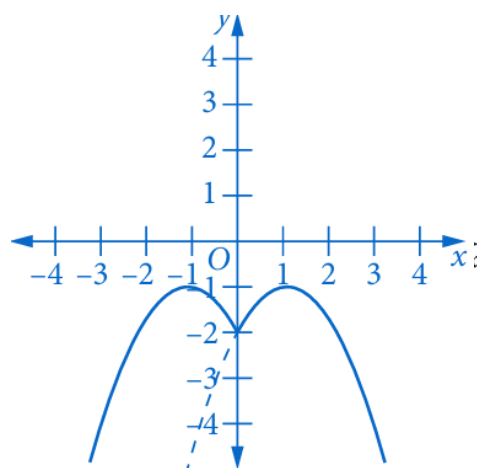
j Sketch $y = f(|x|)$



k Sketch $y = |f(x)|$

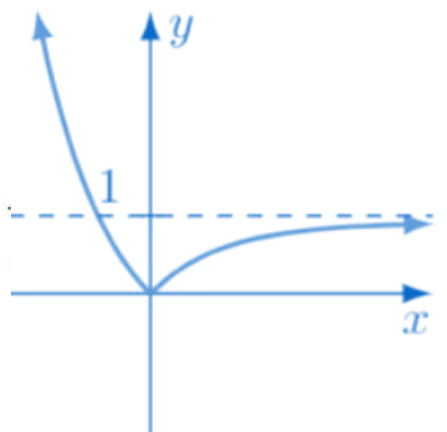


l Sketch $y = f(|x|)$

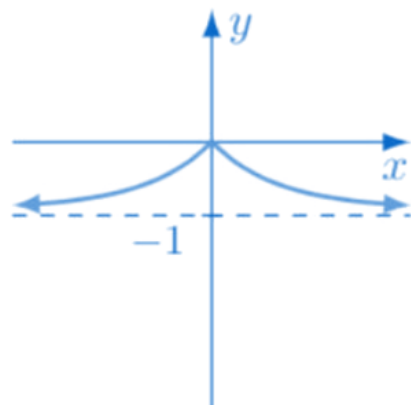


5 The graph of $f(x)$ is given,

a Sketch $y = |f(x)|$

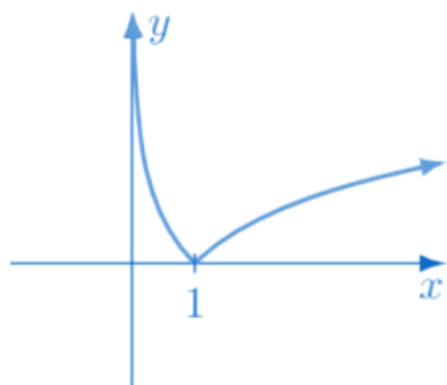


b Sketch $y = f(|x|)$

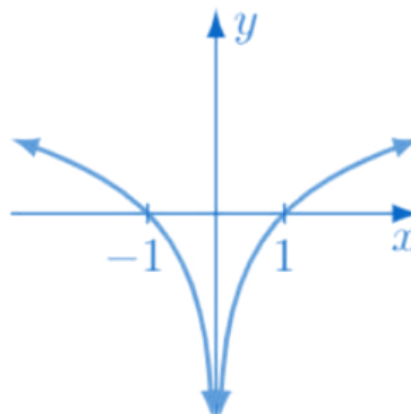


6 The graph of $f(x)$ is given,

a Sketch $y = |f(x)|$

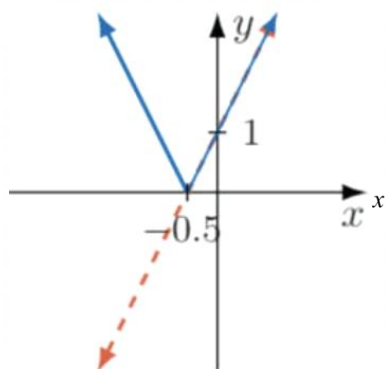


b Sketch $y = f(|x|)$

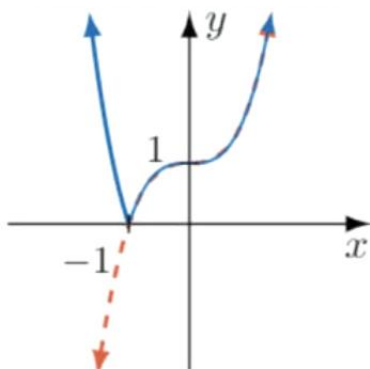


7 By first sketching an appropriate $y = f(x)$, use it to sketch these functions.

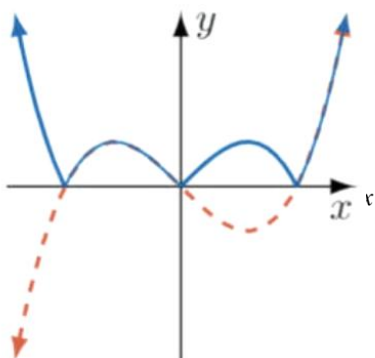
a $y = |2x + 1|$



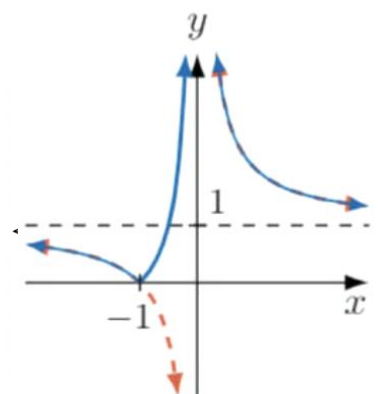
b $y = |x^3 + 1|$



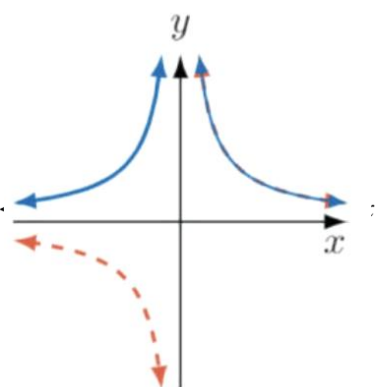
c $y = |x^3 - x|$



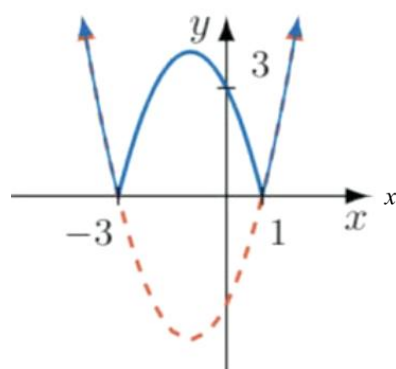
d $y = \left|1 + \frac{1}{x}\right|$



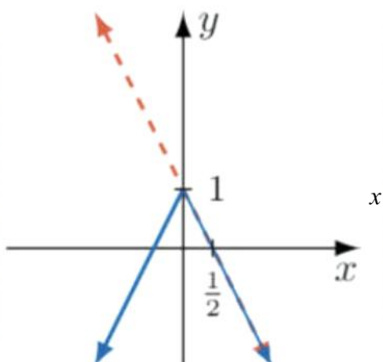
e $y = \frac{1}{|x|}$



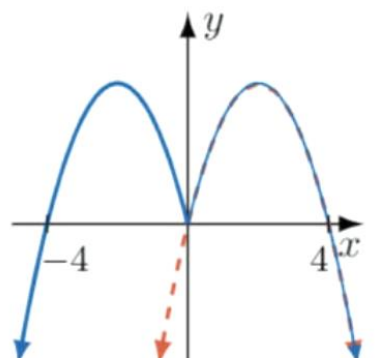
f $y = |x^2 + 2x - 3|$



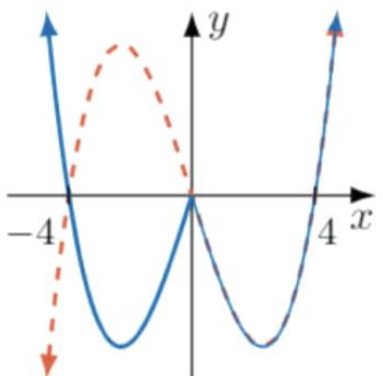
g $y = 1 - 2|x|$



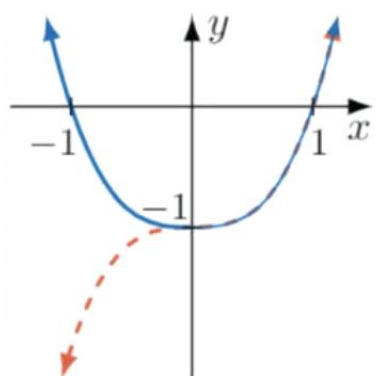
h $y = 4|x| - |x|^2$



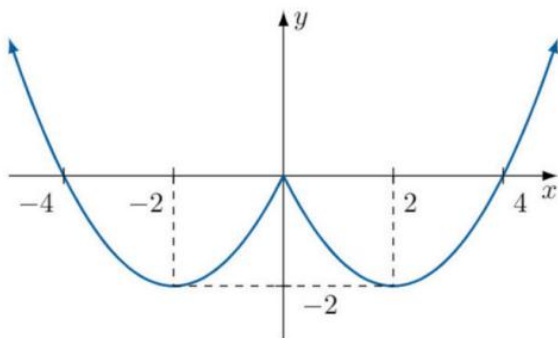
i $y = |x^3| - 16|x|$



j $y = |x|^3 - 1$



- 8 A quadratic was transformed to form the following graph.
Find the equation of the original quadratic.



$$y = \frac{1}{2}(x^2 - 4x)$$

9

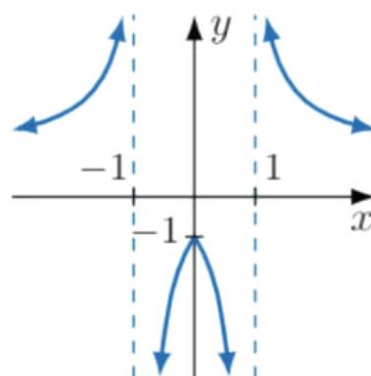
- a Show that

$$\frac{x+1}{x-1} = 1 + \frac{2}{x-1}$$

$$\begin{aligned}\frac{x+1}{x-1} &= \frac{(x-1)+2}{x-1} \\ &= 1 + \frac{2}{x-1}\end{aligned}$$

- b Hence sketch

$$y = \frac{|x|+1}{|x|-1}$$



10

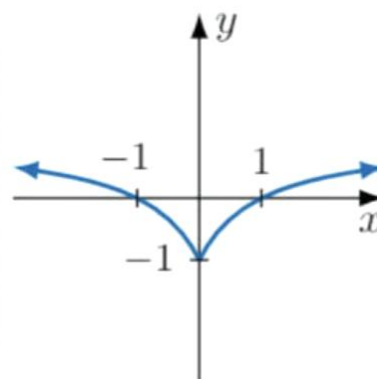
- a Show that

$$\frac{x-1}{x+1} = 1 - \frac{2}{x+1}$$

$$\begin{aligned}\frac{x-1}{x+1} &= \frac{(x+1)-2}{x+1} \\ &= 1 - \frac{2}{x+1}\end{aligned}$$

- b Hence sketch

$$y = \frac{|x|-1}{|x|+1}$$

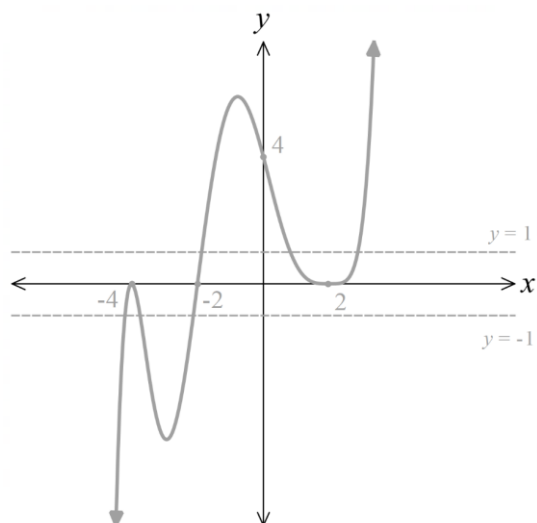


Mixed Practice

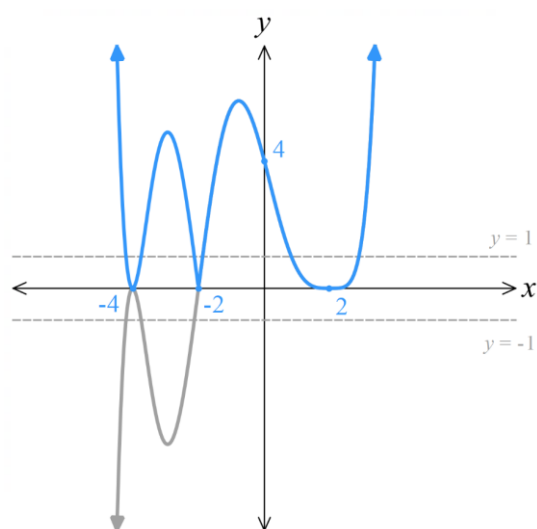
FOUNDATION

1 $y = f(x)$ is given. On the same axis, sketch:

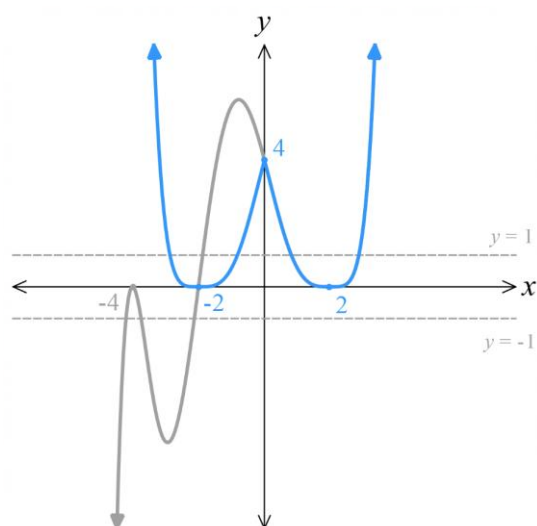
a $y = \frac{1}{f(x)}$



b $y = |f(x)|$

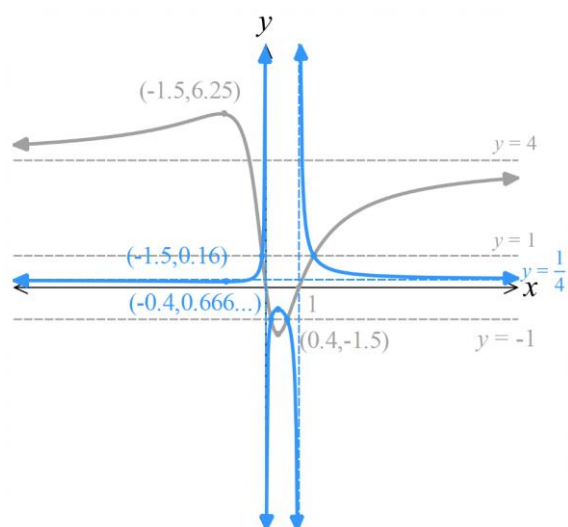


c $y = f(|x|)$

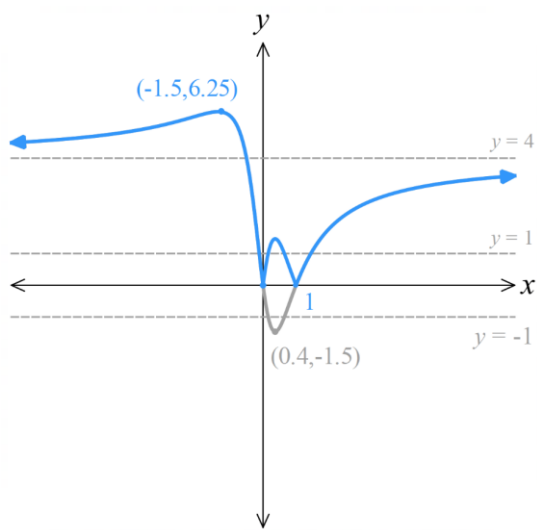


2 $y = f(x)$ is given. On the same axis, sketch:

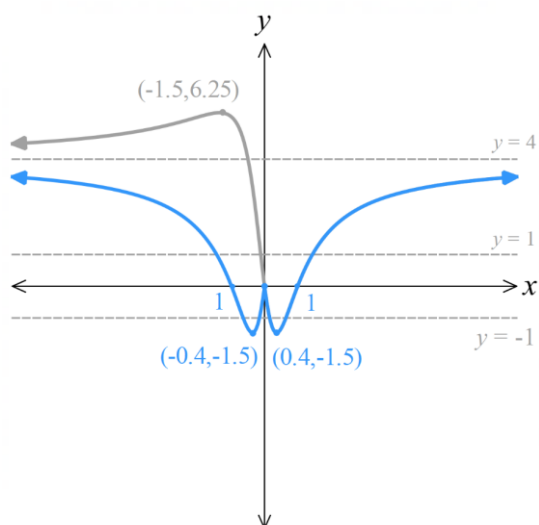
a $y = \frac{1}{f(x)}$



b $y = |f(x)|$

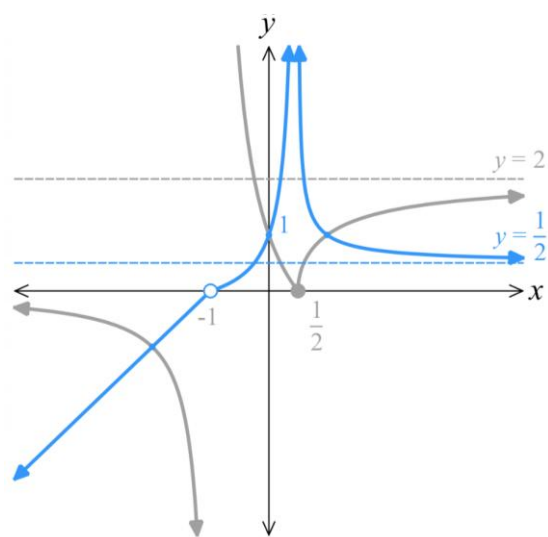


c $y = f(|x|)$

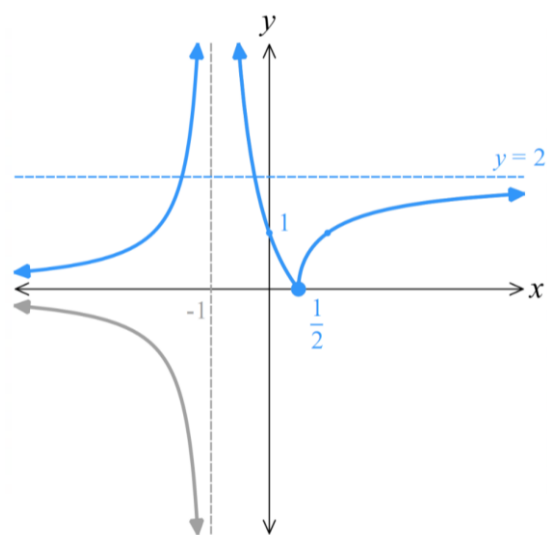


3 $y = f(x)$ is given. On the same axis, sketch:

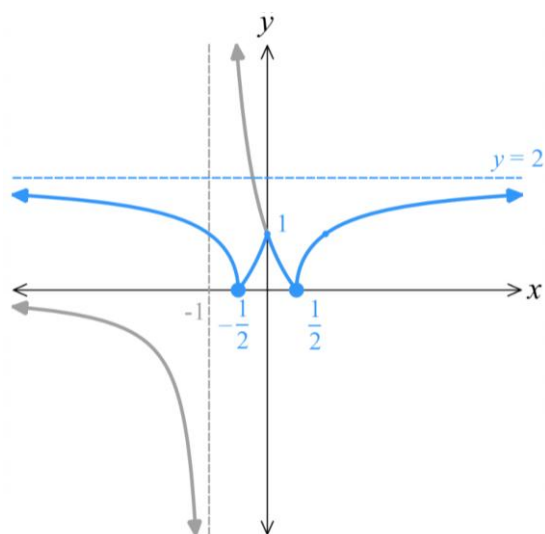
a $y = \frac{1}{f(x)}$



b $y = |f(x)|$



c $y = f(|x|)$



Sum and Difference of Graphs

① Sum of Graphs

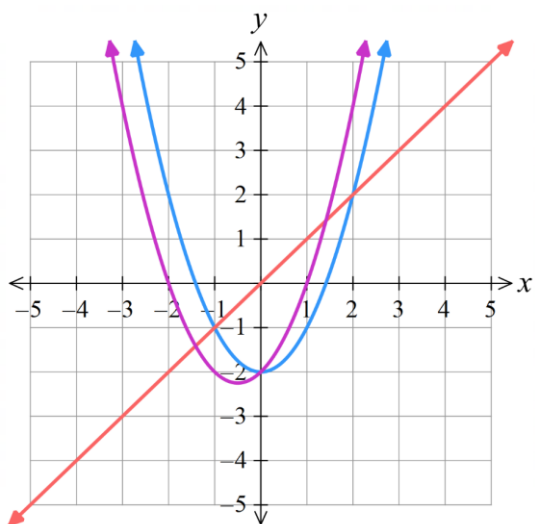
To **add** two graphs $s(x) = f(x) + g(x)$:

- If the graphs are given, add the y -coordinates, wherever possible.
- If the equations of $f(x)$ and $g(x)$ are given, you can add the two functions algebraically to sketch the new function.

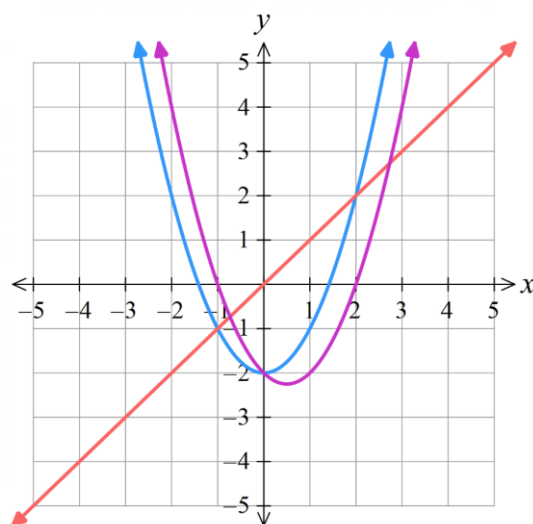
We will focus on the method of adding the **y -coordinates**, also called **ordinates**.

🔍 Sketching the Sum and Difference of Two Graphs

The graphs of $f(x)$ and $g(x)$ are given.
Sketch $s(x) = f(x) + g(x)$.



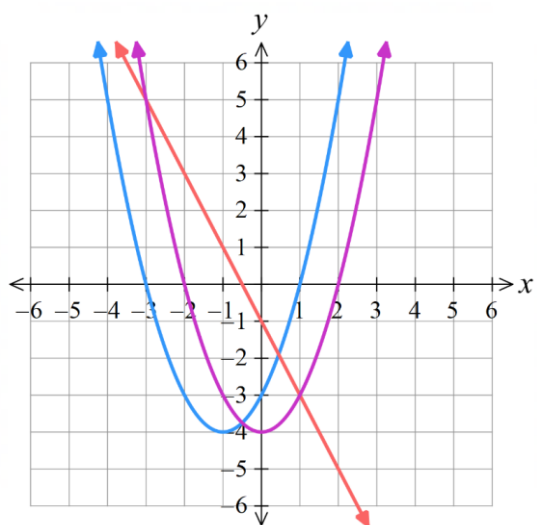
The graphs of $f(x)$ and $g(x)$ are given.
Sketch $d(x) = f(x) - g(x)$.



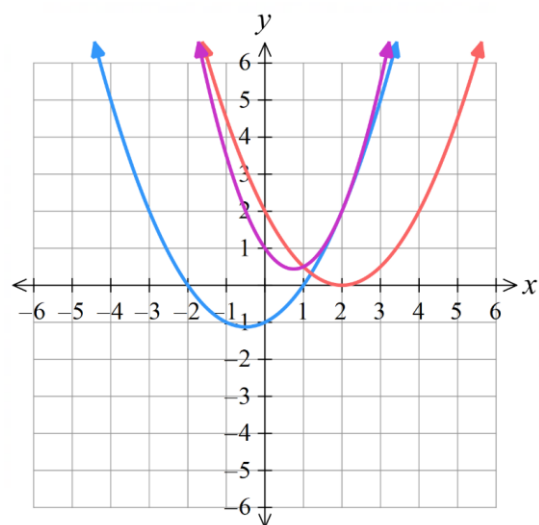
Guided Practice

The graphs of $f(x)$ and $g(x)$ are given. Sketch $s(x) = f(x) + g(x)$.

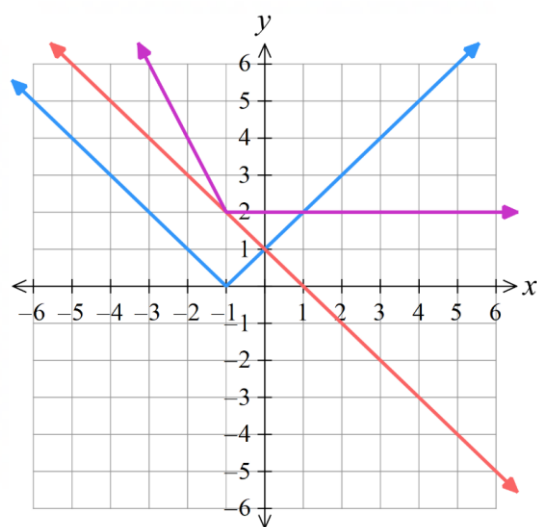
a



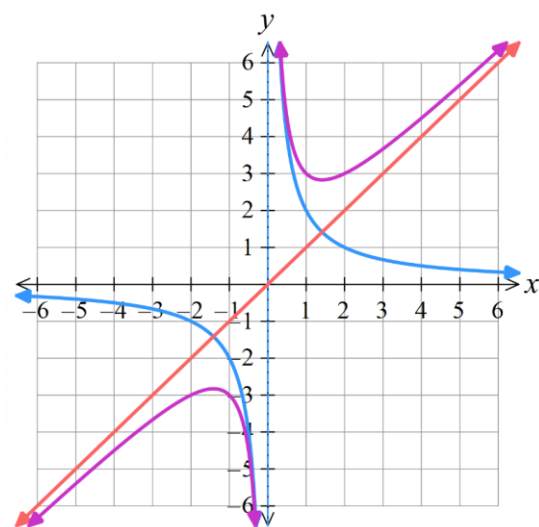
b



c



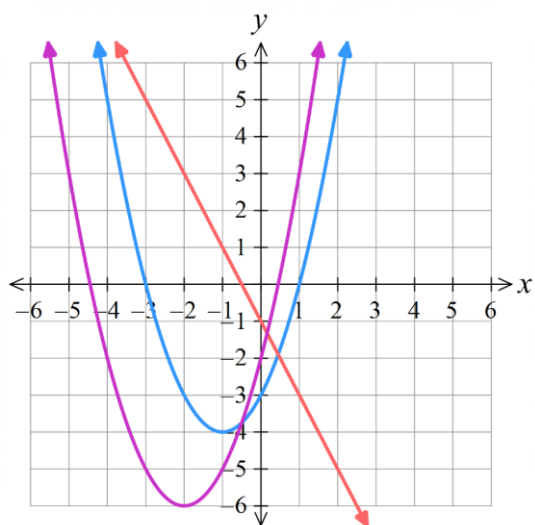
d



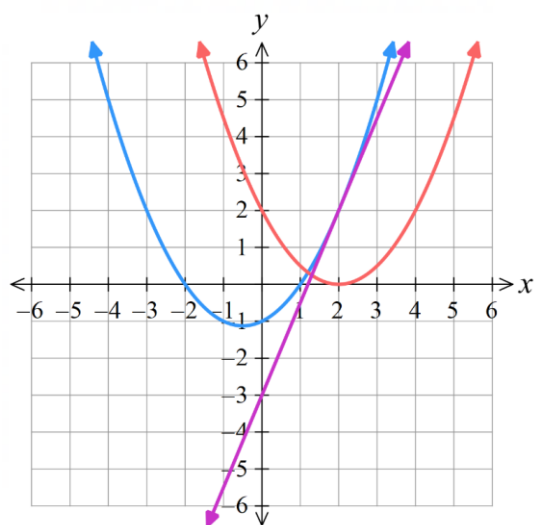
Guided Practice

The graphs of $f(x)$ and $g(x)$ are given. Sketch $d(x) = f(x) - g(x)$.

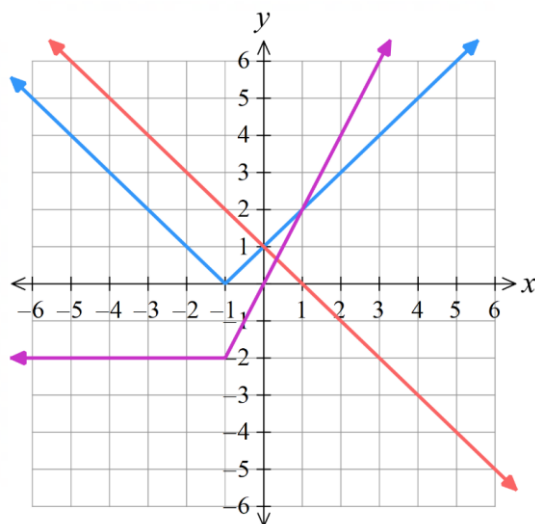
a



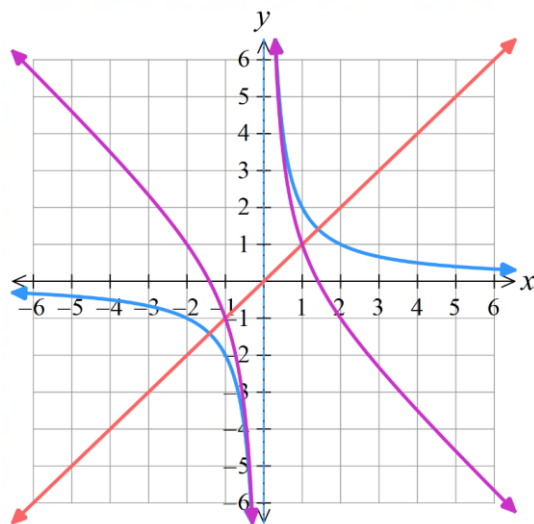
b



c

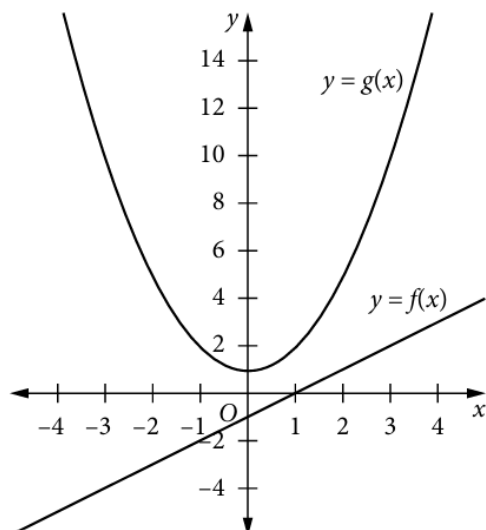


d



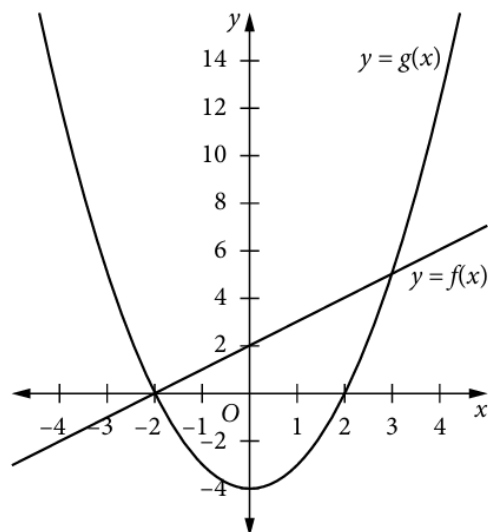
1 By drawing adding ordinates, draw the curve of $y = f(x) + g(x)$.

a



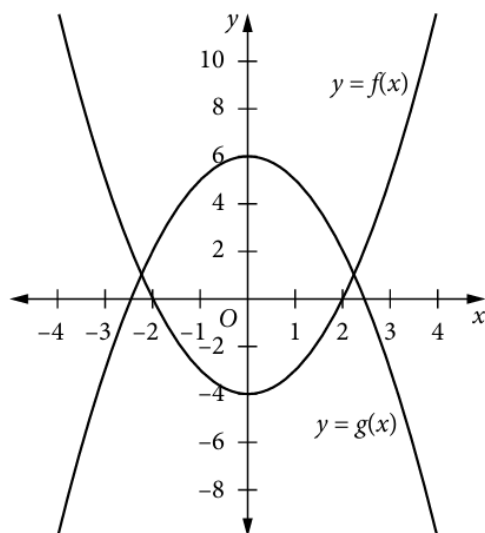
parabola, concave up cutting the x-axis at $(-1, 0)$ and $(0, 0)$

b



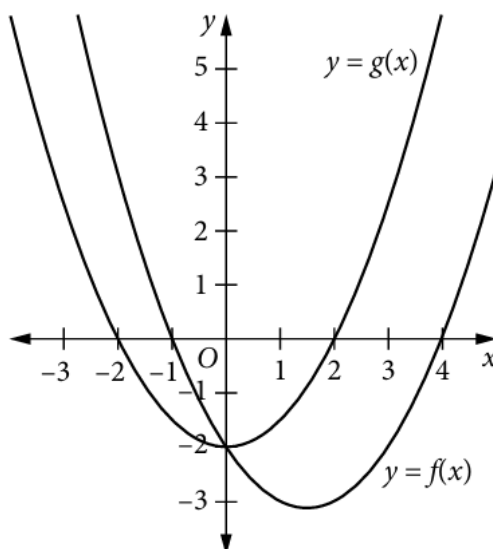
parabola, concave up cutting x-axis at $(-2, 0)$ and $(1, 0)$ and y-axis at $(0, -3)$

c



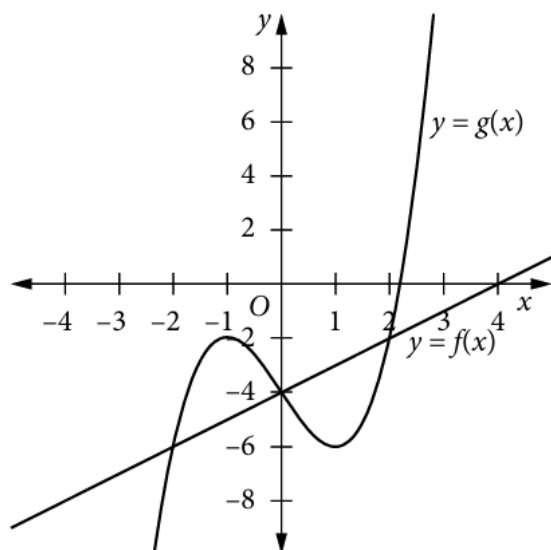
straight line with equation $y = 2$

d



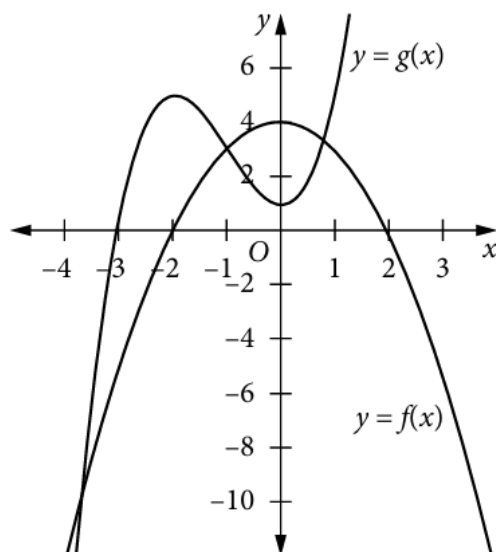
parabola, concave up cutting the y-axis at $(0, -4)$

e



cubic, cutting the x-axis at about $(2.5, 0)$ and the y-axis at $(0, -8)$.
It has the same shape as $y = g(x)$, but lower.

f



cubic, cutting x-axis at about $(-2.6, 0)$ and y-axis at $(0, 5)$.
It has a similar shape to $y = g(x)$, but flattens out more between turning points.

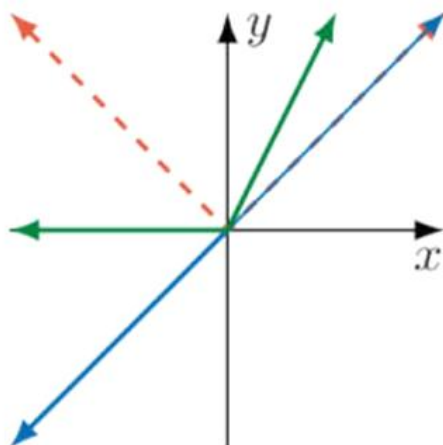
In this section, you will sketch $f(x)$ and $g(x)$ then find the sum or difference.

Show the x & y intercepts and the intersection points of the graphs before adding/subtracting.

2 Sketch the graph of $f(x)$ and $g(x)$ on the same axes, then graph $f(x) + g(x)$.

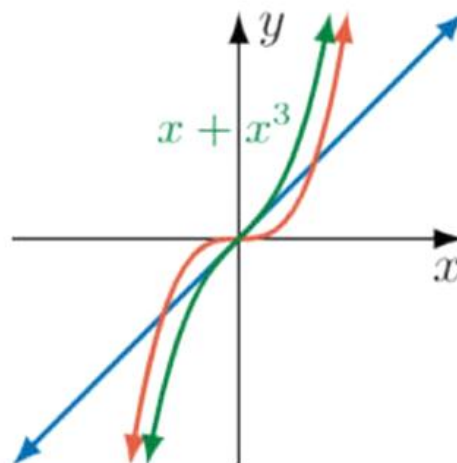
a $f(x) = x$, $g(x) = |x|$

Sketch $y = x + |x|$



b $f(x) = x$, $g(x) = x^3$

Sketch $y = x + x^3$

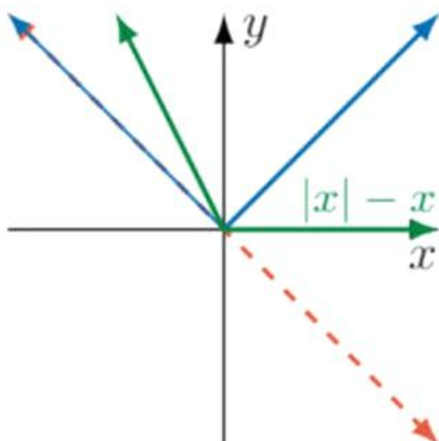


3 Subtracting negatives often leads to mistakes.

Instead, it is easier to subtract graphs by sketching the negative and adding.

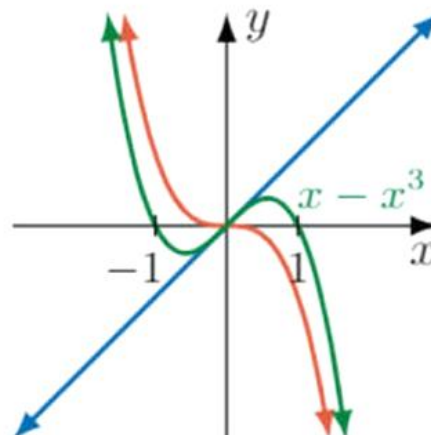
a $f(x) = |x|$, $g(x) = -x$

Sketch $y = |x| - x$



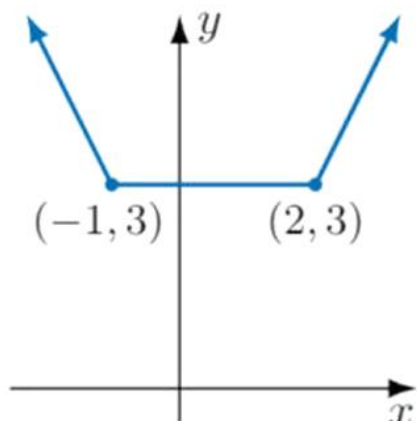
b $f(x) = x$, $g(x) = -x^3$

Sketch $y = x - x^3$

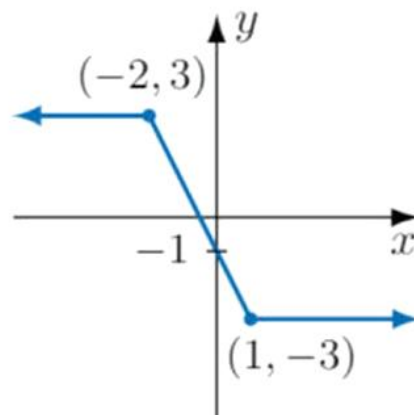


- 4 Sketch these absolute value sums and differences. As absolute value graphs are made up of linear sections, your sums and differences should be made up of straight-line sections.

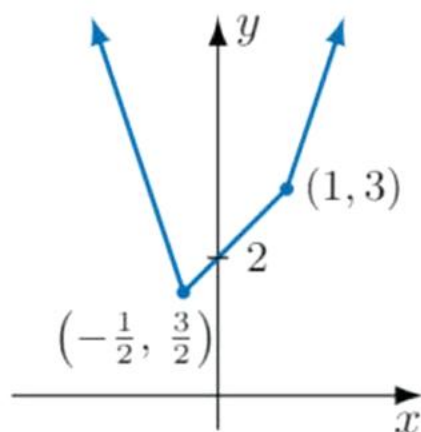
a $y = |x + 1| + |x - 2|$



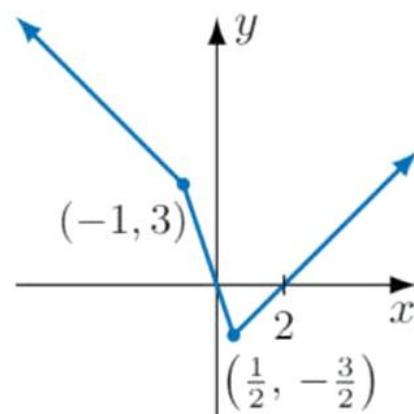
b $y = |x - 1| - |x + 2|$



c $y = |2x + 1| + |x - 1|$

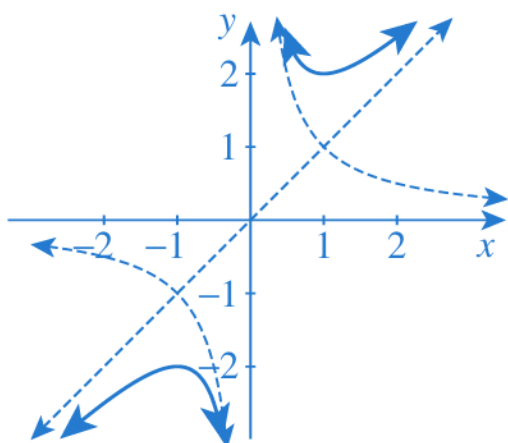


d $y = |2x - 1| - |x + 1|$

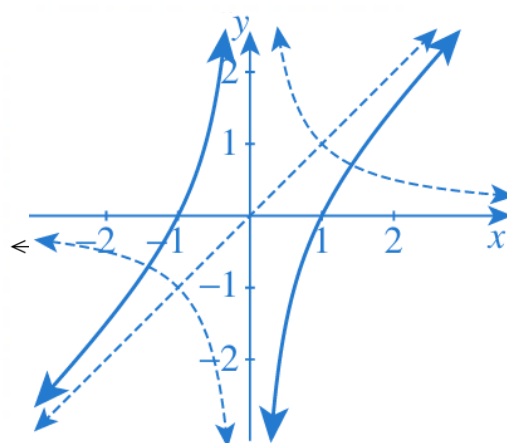


5 Sketch:

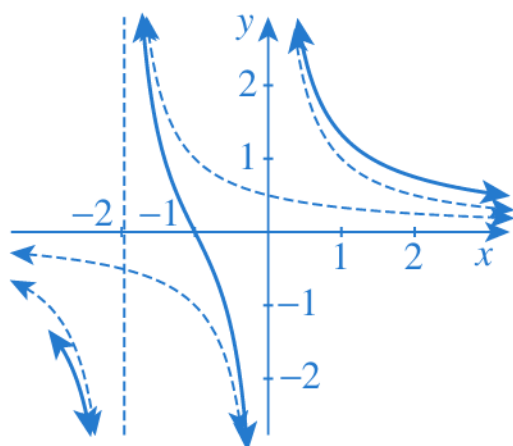
a $y = x + \frac{1}{x}$



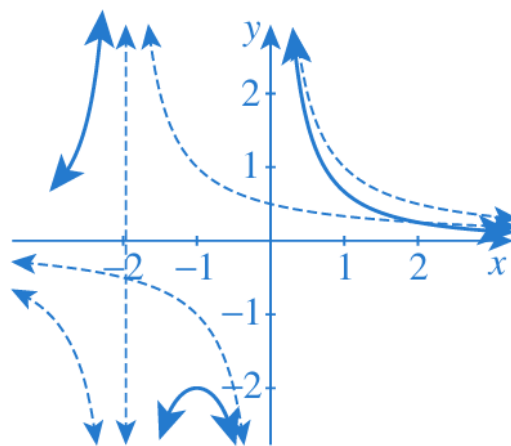
b $y = x - \frac{1}{x}$



c $y = \frac{1}{x} + \frac{1}{x+2}$



d $y = \frac{1}{x} - \frac{1}{x+2}$

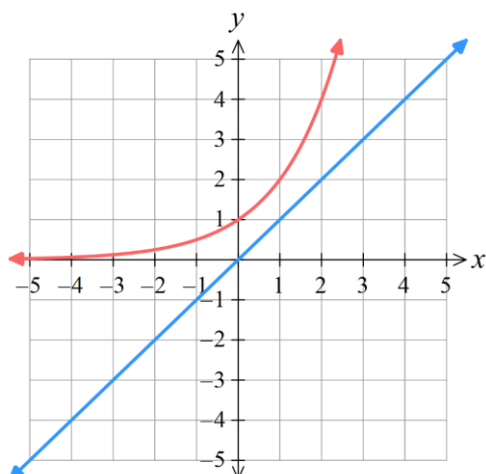


6 Using DESMOS, sketch $y = x$ and $y = 2^x$ and $y = x + 2^x$ on the same axes.

- a What do you notice about the graph of $y = x$ and $y = x + 2^x$ as $x \rightarrow -\infty$?
This is called an oblique asymptote. Explain your observation.

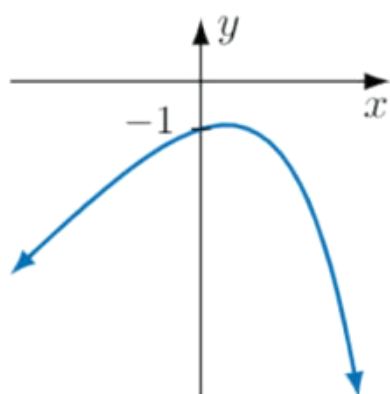
$y = x$ is an asymptote. This is because as $x \rightarrow -\infty$, $2^x \rightarrow 0$, so $x + 2^x \rightarrow x$

- b Attempt to recreate $y = x + 2^x$ by adding ordinates.

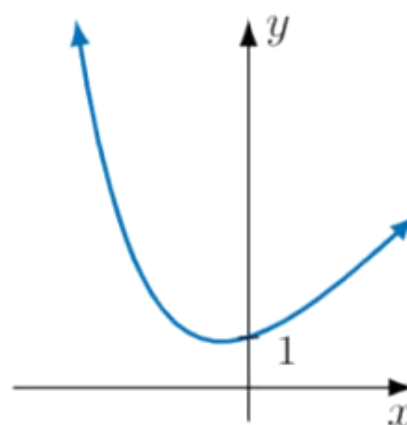


7 Sketch these graphs.

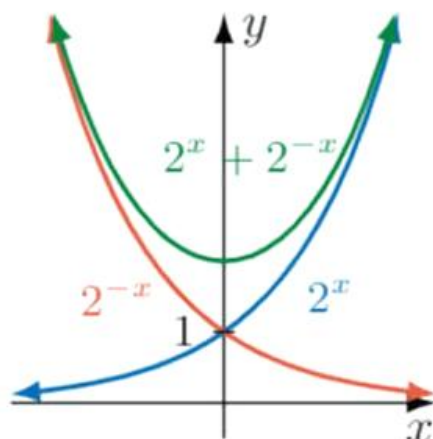
a $y = x - 2^x$



b $y = x + 2^{-x}$



8 Sketch $y = 2^x + 2^{-x}$.



9 Sketch $y = x - \sqrt{x}$.

